

Variational Contraction Conditions for Iterative Algorithms in Multi-Population Discrete-Time Regularized Mean-Field Games*

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Abstract

In this work, we study the contraction conditions of iterative algorithms for stationary and finite-horizon discrete-time regularized mean-field games (MFGs) with multiple populations, where each population only interacts with the state distributions of the other populations. Due to the high dimensionality caused by the interaction of different populations, contraction rates for these algorithms cannot, in general, be expressed in terms of radicals. By studying the dynamics of these iterative algorithms and assuming that the system components of each population's MFG are Lipschitz continuous, we present explicit (eventual) contraction conditions for each algorithm in any normed space, relying only on these Lipschitz parameters. As a consequence of these contraction conditions, we provide convergence rates of finite-horizon mean-field equilibria to infinite-horizon stationary (and nonstationary) mean-field equilibria (MFEs), under restrictions on a variational characterization of the dynamics of these iterative algorithms. In the single-population case, the restrictions we impose on this variational characterization to obtain these convergence results are less restrictive than previous results in the literature.

1 Introduction

In this work, we study the convergence of some basic iterative algorithms in discrete-time regularized mean-field games with multiple populations. We assume that interactions between populations are weak, in the sense that populations interact only through their state distributions, which is a standard setting in the literature on multi-population MFGs [17, 20, 22]. Our main goal is to identify the dynamical systems associated with these iterative algorithms and to study their contraction properties, which imply global asymptotic stability of the mean-field equilibrium (MFE). Beyond deterministic analyses, these global asymptotic stability results also enable the use of basic learning algorithms when components of the MFG are unknown, such as stochastic approximation [9].

In model-based learning settings [15, 19], most methods are restricted to finite-horizon problems. In the single-population case, [6, Theorem 4] shows that learning a finite-horizon MFG with a sufficiently large horizon length can also be sufficient for the infinite-horizon setting, under additional assumptions on the system [6, Corollaries 5 and 6]. Another application of [6, Theorem 4] is a uniqueness condition for infinite-horizon, non-stationary MFEs obtained via finite-horizon MFEs, which is not directly available if one studies the infinite-horizon system itself (see [6, Proposition 3]). For these reasons, it is desirable to obtain contraction conditions in the multi-population setting, particularly for the finite-horizon case, since each of these results relies on establishing a horizon-length-independent contraction condition for finite-horizon MFGs.

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The main difficulty in establishing suitable contraction conditions for multi-population MFGs is the presence of cross-interactions between populations. This cross-interaction, especially in the finite-horizon case, results in a high-dimensional dynamical system, which complicates the analysis. In our case, to study the contraction of this high-dimensional system under Lipschitz continuity assumptions, we will identify a matrix whose spectral radius (which will be an eigenvalue in this case) reveals the contraction property of the corresponding MFG. Since eigenvalues of matrices are roots of their characteristic equations, closed-form expressions for them are often not available in radicals, by a known theorem due to Abel and Ruffini. Furthermore, the matrices that appear in the finite-horizon case will be dense with no structure; thus, the runtime of brute-force calculation of such eigenvalues also scales with the dimension of the matrix, leading to calculating the spectral radius of an $N(T-1) \times N(T-1)$ dimensional matrix when there are N populations and the horizon length is T . Moreover, due to the cross-interaction of these populations, the proofs of the results obtained in the single-population case do not directly scale to the multi-population setting. Thus, easily verifiable contraction conditions are desirable in the multi-population setting.

1.1 Literature Review

This work mainly extends the results of [6] to the multi-population setting. The main distinction between our work and [6] is that the proofs presented in [6] do not scale to the multi-population setting. The contraction condition obtained in [6, Theorem 2] relies on proofs that are purely algebraic (except [6, Lemma 9], which also admits a purely algebraic proof in the setting of that work). In this work, we will transfer the eigenvector problem associated with the matrix that characterizes contraction to the complex plane by using generating functions, and we will argue using complex-analytic methods that are common in the literature on analytic combinatorics [12, Section VI]. For continuous time MFGs, approximation of infinite-horizon MFE via finite-horizon is studied in [10] for the probabilistic solutions.

In discrete-time multi-population MFGs, there is very little work available. For example, [22] studies basic RL algorithms in the setting of LQR average-cost MFGs in discrete time. Although contraction is used to analyze the learning algorithms, [22, Theorem 3] does not provide an easily verifiable condition under which the proposed contraction condition holds. In [20], the existence of an MFE is studied for multi-population discrete-time MFGs in the general case. The work [21] applies model-free learning to stationary multi-population discrete-time MFGs that arise in networks. The work [1] also models a network problem as a multi-population discrete-time average-cost MFG and uses supervised learning to learn the MFE, under a blanket contraction assumption. In [17], a multi-population MFG model is proposed to study epidemics in continuous time, where contraction arguments are used for learning the MFE. We note that all these works also consider weak interactions between populations.

Multi-time-scale analysis of iterative algorithms has also been studied in the discrete-time MFG setting [5]. However, in that work, the global stability conditions required for convergence are justified solely in terms of the global Lipschitz properties of the MFG. In contrast, in Section 6, we show that, even in the multi-population setting, multi-time-scale analysis alone does not improve global contraction properties when one relies only on the Lipschitz properties of the dynamical system.

1.2 Contributions

In this paper, we present quantitative contraction conditions for multi-population stationary and finite-horizon MFGs in the presence of a regularizer.

1. In Section 3, we present a contraction condition for a quasi-static iteration algorithm for multi-population stationary MFGs under regularization (Theorem 2), where one solves an MDP as an intermediate step. This result, in particular, extends [3, Theorem 1] to the multi-population setting.

2. In Section 4, we present *variational contraction conditions* for multi-population finite-horizon MFGs under regularization (Theorem 4), which is the main result of this work. The variational contraction condition that we provide reduces the problem of finding the spectral radius of the matrix to a one-dimensional optimization problem. In particular cases, for instance in the single-population case, the contraction result that we establish is equivalent to [6, Theorem 2]. Moreover, Theorem 4 extends [6, Theorem 2] to the multi-population setting.
3. In Section 5, we present convergence rates and finite-time error bounds between finite-horizon MFEs and infinite-horizon MFEs (Theorem 6). The variational contraction condition that we establish allows us to obtain convergence-rate results even in cases not covered by [6, Theorem 4] in the single-population setting. In particular, our variational contraction condition enables us to establish a convergence rate between finite-horizon and infinite-horizon MFEs whenever the stationary MFE is also contractive, which is not always possible under [6, Theorem 4].
4. In Section 6, we show that the contraction conditions established in Sections 4 and 5 cannot be improved by allowing some populations to update their iterates on faster time scales than others when the only available information about the system consists of the Lipschitz constants of its coefficients; see Corollaries 4, 5, and 6. More precisely, given a family of Lipschitz maps $(f_j)_{j=1}^N$, with $f_j : \prod_{i=1}^N X_i \rightarrow X_j$, we show that any contraction condition for slow-fast updates of the family $(f_j)_{j=1}^N$ derived solely from these Lipschitz constants is equivalent to the eventual contractivity condition for the direct update scheme associated with $(f_j)_{j=1}^N$.

2 Preliminaries

In the mean-field setting, a *population* is a continuum of homogeneous players. In our multi-population mean-field setting, we assume that there is a large number of players, each of whom belongs to a specific population. Each population has its own reward function and transition kernel.

Formally, a discrete-time *multi-population mean-field game (MFG)* is described as follows:

1. The game is played in discrete time, $t \in \{1, 2, \dots\}$.
2. The game is played by a continuum of players divided into N populations. For the sake of simplicity, we assume that each population shares the same finite state space X and finite action space A .
3. A family of state measures $\tau_t = (\tau_{t,1}, \tau_{t,2}, \dots, \tau_{t,N}) \in \prod_{j=1}^N \mathcal{P}(X)$ denotes the global state of the game at time t . The i -th component $\tau_{t,i}$ describes the state distribution of population i . We assume that at each stage t , each player knows their state, their population's state measure, and the state measures of the other populations. Thus, the interaction between populations is *weak*.
4. The one-stage cost of an individual from population i is given by a function $c_i : X \times A \times \prod_{j=1}^N \mathcal{P}(X) \rightarrow [0, \infty)$. The value $c_i(x, a, \tau)$ is the cost incurred at any stage when the player is in state x , takes action a , and the joint state distribution of all populations is τ .
5. State transitions for an individual in population i are defined by the stochastic kernel $p_i : X \times A \times \prod_{j=1}^N \mathcal{P}(X) \rightarrow \mathcal{P}(X)$.

With the conventions above, we represent a multi-population MFG by the tuple

$$(X, A, (c_i)_{i \in [N]}, (p_i)_{i \in [N]}, N),$$

so that a multi-population MFG can be viewed as a collection of N distinct discrete-time MFGs. The main distinction between a multi-population MFG and a single-population MFG is the cross-interaction between different populations.

For a given $t \in \mathbb{N}$ and $i \in [N]$, a *policy* is a stochastic kernel $\pi_{t,i} : X \rightarrow \mathcal{P}(A)$ that specifies the distribution over actions for population i at time t given a state. We denote by Π the space of policies $(\pi_{t,i})_{t \in \mathbb{N}, i \in [N]}$. Policies being independent of the state measures of other populations is an artifact of the weak interaction assumption.

In general, a discrete-time MFG is an extension of discrete-time Markov decision processes (MDPs) [18]. Indeed, a MFG can be reduced to an MDP if one operates under a fixed family of state measures $(\tau_t)_{t=0}^\infty$. Let $(\beta_i)_{i \in [N]} \subset (0, 1)$. Thus, in the infinite-horizon case, to define an objective function for each population, we need additional structure on the MFG. For each $i \in [N]$, $\beta_i < 1$ is the discount factor used by population i . Under these discount factors, and for a given family of multi-population state measures $\boldsymbol{\tau} = (\tau_t)_{t=0}^\infty$, the discounted objective function of the population $i \in [N]$ is defined as

$$J_i(\boldsymbol{\pi}, \boldsymbol{\tau}) = \mathbb{E}_{\boldsymbol{\pi}} \left[\sum_{t=0}^{\infty} \beta_i^t c_i(x_t, a_t, \tau_{t,1}, \tau_{t,2}, \dots, \tau_{t,N}) \right],$$

where $\boldsymbol{\pi} = (\pi_{t,i})_{t \in \mathbb{N}, i \in [N]}$. For the objective of the population $i \in [N]$, the evolutions of the states and actions are given by

$$x(0) \sim \tau_{0,i}, \quad x(t) \sim p_i(\cdot | x(t-1), a(t-1), \tau_{t,i}), \quad t \geq 1, \quad a(t) \sim \pi_{t,i}(\cdot | x(t)), \quad t \geq 0.$$

We say that a family of policies $\boldsymbol{\pi} := (\pi_{t,i})_{t \in \mathbb{N}, i \in [N]}$ is *optimal* for a family of state measures $\boldsymbol{\tau} := (\tau_{t,i})_{t \in \mathbb{N}, i \in [N]}$ if, for all $i \in [N]$,

$$\tau_{t+1,i}(\cdot) = \int_X \int_A p_i(\cdot | x, a, \tau_{t,1}, \tau_{t,2}, \dots, \tau_{t,N}) \pi_{t,i}(da | x) \tau_{t,i}(dx). \quad (1)$$

Conversely, we say that a family of state measures $\boldsymbol{\tau}$ is *optimal* for a family of policies $\boldsymbol{\pi}$ if, for all $i \in [N]$,

$$J_i(\boldsymbol{\pi}, \boldsymbol{\tau}) = \inf_{\tilde{\boldsymbol{\pi}} \in \Pi} J_i(\tilde{\boldsymbol{\pi}}, \boldsymbol{\tau}).$$

A solution to the MFG consists of a family of policies $(\pi_{t,i})_{t \in \mathbb{N}, i \in [N]}$ and state measures $(\tau_{t,i})_{t \in \mathbb{N}, i \in [N]}$ such that $(\pi_{t,i})_{t \in \mathbb{N}, i \in [N]}$ is optimal for $(\tau_{t,i})_{t \in \mathbb{N}, i \in [N]}$ and $(\tau_{t,i})_{t \in \mathbb{N}, i \in [N]}$ is optimal for $(\pi_{t,i})_{t \in \mathbb{N}, i \in [N]}$. In this case, we call $((\pi_{t,i})_{t \in \mathbb{N}, i \in [N]}, (\tau_{t,i})_{t \in \mathbb{N}, i \in [N]})$ a *mean-field equilibrium* (MFE).

In this work, we are interested in computing MFEs through iterative algorithms. For this purpose, we will mainly work with *deterministic policies*. A policy $\boldsymbol{\pi} = (\pi_{t,i})_{t \in \mathbb{N}, i \in [N]} \in \Pi$ is said to be deterministic if, for all $t \in \mathbb{N}$ and $i \in [N]$, there exists a function $f_{t,i} : X \rightarrow A$ such that $\pi_{t,i}(\cdot | x) = \delta_{f_{t,i}(x)}(\cdot)$, where δ denotes the Dirac measure. The space of all deterministic policies is then denoted by Π_d .

To compute an MFE, throughout the iterations, we will need to ensure that there exists a unique Lipschitz policy. To achieve this, we first lift the action space of the MFG to $\mathcal{P}(A)$. This amounts to defining an MFG $(X, \mathcal{P}(A), (C_i)_{i \in [N]}, (P_i)_{i \in [N]}, N)$ that shares the “same” MFE as $(X, A, (c_i)_{i \in [N]}, (p_i)_{i \in [N]}, N)$. First, we embed $\mathcal{P}(A) \subset \mathbb{R}^{|A|}$. For any $u \in \mathcal{P}(A)$, we define

$$C_i(x, u, \tau) = C_i(x, u, \tau_1, \dots, \tau_N) = \sum_{a \in A} c_i(x, a, \tau_1, \dots, \tau_N) u(a),$$

and

$$P_i(\cdot | x, u, \tau) = P_i(\cdot | x, u, \tau_1, \dots, \tau_N) = \sum_{a \in A} p_i(\cdot | x, a, \tau_1, \dots, \tau_N) u(a).$$

Now, to ensure that we obtain unique Lipschitz policies during our iterations, we perturb each C_i as $C_i + \rho_i \Omega_i$, where $\rho_i > 0$ and $\Omega_i : \mathcal{P}(A) \rightarrow \mathbb{R}$. We further assume that each Ω_i is 1-strongly

convex with respect to the ℓ_1 -norm, i.e.,

$$\Omega_i(\theta u + (1 - \theta)v) \leq \theta \Omega_i(u) + (1 - \theta) \Omega_i(v) - \frac{1}{2} \theta(1 - \theta) \|u - v\|_1^2$$

for all $u, v \in \mathcal{P}(A)$ and $\theta \in [0, 1]$. The resulting MFG $(X, \mathcal{P}(A), (C_i + \rho_i \Omega_i)_{i \in [N]}, (P_i)_{i \in [N]}, N)$ is called a *regularized MFG*, and the parameters ρ_i are called *regularization parameters*. The MFE obtained from the regularized MFG deviates from that of the original lifted MFG; however, these perturbations ensure the existence of a unique Lipschitz flow of policies that achieves the objective function for any given family of state measures. We define a candidate domain for our Q -functions at the multi-population level as follows: Let M_i be a uniform bound for the cost function c_i . The candidate space of Q -functions for the population i over the action space A is

$$\tilde{\mathcal{C}}_i := \left\{ Q^{\text{disc}} : X \times A \rightarrow \mathbb{R}_+ : |Q^{\text{disc}}(x, a) - Q^{\text{disc}}(\tilde{x}, \tilde{a})| \leq \frac{L_i}{1 - \frac{\beta_i K_i}{2}} (1_{x \neq \tilde{x}} + 2 1_{a \neq \tilde{a}}), \|Q\|_\infty \leq \frac{M_i}{1 - \beta_i} \right\},$$

and then we lift them to $\mathcal{P}(A)$ and consider the perturbation with a regularizer Ω_i as follows:

$$\mathcal{C}_i := \left\{ Q : X \times \mathcal{P}(A) \rightarrow \mathbb{R} : Q(x, u) = \sum_{a \in A} Q^{\text{disc}}(x, a) u(a) + \rho_i \Omega_i(u), \|Q\|_\infty \leq \frac{M_i + \|\rho_i \Omega_i\|_\infty}{1 - \beta_i}, Q^{\text{disc}} \in \tilde{\mathcal{C}}_i \right\}.$$

The algorithms that we consider in this work will closely mimic the classical value-iteration algorithm for MDPs. For a given family of state measures $\boldsymbol{\tau}$, we update our Q -functions as in value iteration, and, to obtain the consistency condition (1), we also need to control the updates of our state measures. For this purpose, we define the following operators:

Definition 1. Let $Q \in \mathcal{C}_i$ and $\tau = (\tau_i)_{i=1}^N \in \prod_{i=1}^N \mathcal{P}(X)$. For updating our Q -functions, we define the operator

$$H_{1,i}(Q, \tau)(x, u) = C_i(x, u, \tau) + \rho_i \Omega_i(u) + \beta_i \sum_{y \in X} Q_{\min}(y) P_i(y | x, u, \tau),$$

and for updating our state measures, we define

$$H_{2,i}(Q, \tau)(\cdot) = \sum_{x \in X} \int_{\mathcal{P}(A)} P_i(\cdot | x, u, \tau) \pi_Q(du | x) \tau_i(x),$$

where π_Q is an optimal policy induced by the Q -function Q .

The operator $H_{1,i}$ arises as a consequence of the dynamic programming structure associated with the objective functions J_i , a well-known result in the theory of MDPs. The operator $H_{2,i}$ is specific to the MFG setting and arises from the consistency requirement linking policies and state measures through cross-population interactions. From the definition of $H_{1,i}$, note that since C_i and P_i are affine over $\mathcal{P}(A)$ and Ω_i is strongly convex, for any fixed Q -function there exists a unique minimizer of $H_{1,i}(Q, \tau)(x, \cdot)$ for each (x, τ) . In particular, the operator $H_{2,i}$ is well defined.

In this work, we will deal with three different types of MFE. In the infinite-horizon case, an MFE $(\boldsymbol{\pi}, \boldsymbol{\tau})$ is said to be *stationary* if it is time independent, i.e., $\tau_{t,i} = \tau_{t+1,i}$ and $\pi_{t,i} = \pi_{t+1,i}$ for all $i \in [N]$ and all $t \in \mathbb{N}$. Due to this stationarity condition, for a stationary MFE we do not impose any restriction on the initial state measure. If the initial state measures are not fixed, one often seeks a stationary MFE.

Given a family of initial state measures $\tau_0 = (\tau_{0,i})_{i \in [N]} \in \prod_{i \in [N]} \mathcal{P}(X)$, we say that an MFE $(\boldsymbol{\pi}, (\tilde{\tau}_{t,i})_{t \in \mathbb{N}, i \in [N]})$ is *non-stationary* if $\tilde{\tau}_{0,i} = \tau_{0,i}$ for all $i \in [N]$. In the non-stationary case, one fixes the initial state measures *a priori*, which usually come from the population's initial distribution. For an initial state measure $\tau_0 \in \prod_{i \in [N]} \mathcal{P}(X)$, we denote by MFE_{τ_0} the set of MFEs for the infinite-horizon MFG starting from τ_0 .

In the finite-horizon case, one often proceeds as in the infinite-horizon case, with the caveat that the objective function takes the form

$$J_i(\boldsymbol{\pi}, \boldsymbol{\tau}) = \mathbb{E}_{\boldsymbol{\pi}} \left[\sum_{t=0}^{T-1} \beta_i^t c_i(x_t, a_t, \tau_{t,1}, \tau_{t,2}, \dots, \tau_{t,N}) \right].$$

The truncation of the sum at time T imposes a natural boundary condition on the iterations of the Q -functions. In this case, we denote by MFE_{T, τ_0} the set of finite-horizon MFEs starting from τ_0 .

To complete our setup, we impose a Lipschitz continuity assumption on the system components c_i and p_i as follows:

Assumption 1. We denote by $\|\cdot\|_1$ the ℓ_1 -norm on $\mathcal{P}(X) \subset \mathbb{R}^{|X|}$. For all $i \in [N]$, $x, \tilde{x} \in X$, $a, \tilde{a} \in \mathcal{P}(A)$, and $\tau = (\tau_i)_{i \in [N]}, \tilde{\tau} = (\tilde{\tau}_i)_{i \in [N]} \in \prod_{i \in [N]} \mathcal{P}(X)$, we have

$$|c_i(x, a, \tau) - c_i(\tilde{x}, \tilde{a}, \tilde{\tau})| \leq L_i \left(\mathbf{1}_{\{x \neq \tilde{x}\}} + 2 \mathbf{1}_{\{a \neq \tilde{a}\}} + \sum_{j=1}^N \|\tau_j - \tilde{\tau}_j\|_1 \right),$$

and

$$\|p_i(\cdot | x, a, \tau) - p_i(\cdot | \tilde{x}, \tilde{a}, \tilde{\tau})\|_1 \leq K_i \left(\mathbf{1}_{\{x \neq \tilde{x}\}} + 2 \mathbf{1}_{\{a \neq \tilde{a}\}} + \sum_{j=1}^N \|\tau_j - \tilde{\tau}_j\|_1 \right).$$

For the rest of the paper, we will suppose that Assumption 1 is effective. Lipschitz continuity assumptions such as the ones above are common in the analysis of iterative algorithms.

Although we have defined the operators $H_{1,i}$ and $H_{2,i}$ for the lifted system, Assumption 1 above is stated for the original system. It turns out, an analogous Lipschitz continuity condition also holds for the lifted system [3].

Lemma 1. For all $x, \tilde{x} \in X$, $u, \tilde{u} \in \mathcal{P}(A)$, and $\tau, \tilde{\tau} \in \prod_{i \in [N]} \mathcal{P}(X)$, C_i and P_i satisfy the following Lipschitz bounds:

$$|C_i(x, u, \tau) - C_i(\tilde{x}, \tilde{u}, \tilde{\tau})| \leq L_i \left(\mathbf{1}_{\{x \neq \tilde{x}\}} + \|u - \tilde{u}\|_1 + \sum_{j \in [N]} \|\tau_j - \tilde{\tau}_j\|_1 \right),$$

and

$$\|P_i(\cdot | x, u, \tau) - P_i(\cdot | \tilde{x}, \tilde{u}, \tilde{\tau})\|_1 \leq K_i \left(\mathbf{1}_{\{x \neq \tilde{x}\}} + \|u - \tilde{u}\|_1 + \sum_{j \in [N]} \|\tau_j - \tilde{\tau}_j\|_1 \right).$$

Proof. The same proof in [3] holds verbatim, so we omit the details. $\square$

An important consequence of Lemma 1 is the following Lipschitz property of $H_{1,i}$ and $H_{2,i}$, which will be fundamental for deriving a variational expression for our iterations in the stationary and finite-horizon settings. We will provide a proof for such an expression by closely following the proof of the single population case presented in [3]. Since our main results will heavily rely on these variational inequalities, we will provide a detailed proof for completeness in the appendix.

Lemma 2. For any $Q, \tilde{Q} \in \mathcal{C}_i$ and $\tau, \tilde{\tau} \in \prod_{i=1}^N \mathcal{P}(X)$ we have

$$\|H_{1,i}(Q, \tau) - H_{1,i}(\tilde{Q}, \tilde{\tau})\|_{\infty} \leq \beta_i \|Q - \tilde{Q}\|_{\infty} + \frac{L_i}{1 - \frac{\beta_i K_i}{2}} \left(\sum_{j=1}^N \|\tau_j - \tilde{\tau}_j\|_1 \right), \quad (2)$$

and

$$\|H_{2,i}(Q, \tau) - H_{2,i}(\tilde{Q}, \tilde{\tau})\|_1 \leq \frac{K_i}{\rho_i} \|Q - \tilde{Q}\|_{\infty} + \bar{K}_i \|\tau_i - \tilde{\tau}_i\|_1 + K_i \sum_{j \neq i} \|\tau_j - \tilde{\tau}_j\|_1, \quad (3)$$

where $\bar{K}_i = \frac{3}{2} K_i + \frac{1}{2} \frac{\bar{L}_i K_i}{\rho_i (1 - \beta_i)}$, and $\bar{L}_i = \frac{L_i}{1 - \frac{\beta_i K_i}{2}}$.

Proof. We postpone the proof to Appendix A. $\square$

We also note that for a given $\tau \in \prod_{i \in [N]} \mathcal{P}(X)$, $H_{1,i}(\cdot, \tau)$ maps $\mathcal{C}_i$ to itself.

Lemma 3. *For each $i \in [N]$, $H_{1,i}(\cdot, \tau)$ maps $\mathcal{C}_i$ to itself.*

Proof. The same arguments in the proofs [6, Lemma 2] and [4, Lemma 1] are directly applicable here; so, we will omit the details. $\square$

3 Eventual Contractivity of a Quasi-Static Iterative Algorithm for Multi-population Discounted Stationary MFGs

Let $(X, \mathcal{P}(A), (C_i + \rho_i \Omega_i)_{i \in [N]}, (P_i)_{i \in [N]}, N)$ be a regularized multi-population MFG, where the objective function of population i is discounted by the factor β_i , for each $i \in [N]$. In this section, we establish a contraction condition to an iterative algorithm to compute the MFE of this multi-population stationary MFG. Since we are looking for stationary equilibria, the multi-population case amounts to dealing with N interacting dynamical systems. Since stationary solutions are time-independent, contraction conditions for stationary MFE induce a simpler structure to analyze compared to the non-stationary setting, in both the finite- and infinite-horizon cases.

The iterative algorithm in the stationary MFE case involves updating both Q -functions (using the current state measures as input) and state measures (using the minimizer induced by the current Q -functions). For simplicity, we will study the quasi-static case, where one finds a fixed point of $H_{1,i}(\cdot, \tau)$ at each step for a given initial state-measure $\tau \in \prod_{i \in [N]} \mathcal{P}(X)$. In practice, when the model is unknown, these Q -functions can be calculated using stochastic approximation [13].

First, to provide a formal definitions of this algorithm, we introduce the following notation:

Definition 2. *For a given $\tau \in \prod_{i=1}^N \mathcal{P}(X)$, let $Q^{\tau,i} \in \mathcal{C}_i$ denote the Q -function satisfying*

$$H_{1,i}(Q^{\tau,i}, \tau) = Q^{\tau,i}. \quad (4)$$

Since $\max_{i \in [N]} \beta_i < 1$, this fixed point exists and is unique for every $i \in [N]$, as a consequence of the contraction property of the Bellman operator. Moreover, $Q^{\tau,i} \in \mathcal{C}_i$ by Lemma 3.

When the population i is clear from the context, we will also refer to $Q^{\tau,i}$ simply as the Q -function corresponding to τ , somewhat abusing the notation. An iterative algorithm that finds a stationary MFE in this case is then the following:

Algorithm 1: Quasi-Static Value Iteration for Multi Population MFG

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Initialize with  $\tau_0 = (\tau_{i,0})_{i \in [N]} \in \prod_{i \in [N]} \mathcal{P}(X)$ ;
while  $\tau_{n+1} \neq \tau_n$  do
  |  $\tau_{n+1} = (H_{2,i}(Q^{\tau_n,i}, \tau_n))_{i \in [N]}$ ;
end
Let  $\pi_i^* = \delta_{\argmin_{u \in \mathcal{P}(A)} Q^{\tau^*,i}(\cdot, u)}(\cdot)$  for all  $i \in [N]$ ;
Let  $\pi^* = (\pi_i^*)_{i \in [N]}$ ;
return stationary mean-field equilibrium  $(\pi^*, \tau^*)$ 

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As described in the algorithm, the fixed points of the map $\tau \mapsto (H_{2,i}(Q^{\tau,i}, \tau))_{i \in [N]}$ induce stationary MFE. To study the convergence of this algorithm, we will provide an (eventual) contraction condition for $\tau \mapsto (H_{2,i}(Q^{\tau,i}, \tau))_{i \in [N]}$.

In the next lemma, we will provide a matrix inequality for the map $(H_{2,i})_{i \in [N]}$ that will be fundamental while obtaining an eventual contraction condition that ensures the convergence of Algorithm 1.

Lemma 4. For all $\tau, \tilde{\tau} \in \prod_{i \in [N]} \mathcal{P}(X)$, we have

$$\begin{bmatrix} \|H_{2,1}(Q^{\tau,1}, \tau) - H_{2,1}(Q^{\tilde{\tau},1}, \tilde{\tau})\|_1 \\ \|H_{2,2}(Q^{\tau,2}, \tau) - H_{2,2}(Q^{\tilde{\tau},2}, \tilde{\tau})\|_1 \\ \vdots \\ \|H_{2,N}(Q^{\tau,N}, \tau) - H_{2,N}(Q^{\tilde{\tau},N}, \tilde{\tau})\|_1 \end{bmatrix} \leq \mathcal{M} \begin{bmatrix} \|\tau_1 - \tilde{\tau}_1\|_1 \\ \|\tau_2 - \tilde{\tau}_2\|_1 \\ \vdots \\ \|\tau_N - \tilde{\tau}_N\|_1 \end{bmatrix},$$

where $\mathcal{M} = [M_{i,j}]_{i,j}$ is such that $M_{i,j} = \begin{cases} \bar{K}_i + \frac{\bar{L}_i K_i}{\rho_i(1 - \beta_i)}, & i = j, \\ K_i + \frac{\bar{L}_i K_i}{\rho_i(1 - \beta_i)}, & j \neq i. \end{cases}$

Proof. For any $j \in [N]$, using Lemma 2 we obtain

$$\|H_{2,j}(Q^{\tau,j}, \tau) - H_{2,j}(Q^{\tilde{\tau},j}, \tilde{\tau})\|_1 \leq \bar{K}_j \|\tau_j - \tilde{\tau}_j\|_1 + \frac{K_j}{\rho_j} \|Q^{\tau,j} - Q^{\tilde{\tau},j}\|_\infty + K_j \left(\sum_{i \neq j} \|\tau_i - \tilde{\tau}_i\|_1 \right).$$

Using the Bellman equation (4) and Lemma 2, it also follows that

$$\|Q^{\tau,j} - Q^{\tilde{\tau},j}\|_\infty \leq \frac{1}{1 - \beta_j} \bar{L}_j \left(\sum_{i \in [N]} \|\tau_i - \tilde{\tau}_i\|_1 \right).$$

Combining these two inequalities, we get

$$\begin{aligned} \|H_{2,j}(Q^{\tau,j}, \tau) - H_{2,j}(Q^{\tilde{\tau},j}, \tilde{\tau})\|_1 &\leq \left(\bar{K}_j + \frac{\bar{L}_j K_j}{\rho_j(1 - \beta_j)} \right) \|\tau_j - \tilde{\tau}_j\|_1 \\ &\quad + \left(K_j + \frac{\bar{L}_j K_j}{\rho_j(1 - \beta_j)} \right) \left(\sum_{i \neq j} \|\tau_i - \tilde{\tau}_i\|_1 \right), \end{aligned}$$

as desired □

By $\rho(\mathcal{M})$, we denote the spectral radius of the matrix $\mathcal{M}$. Let $\|\cdot\|$ be a matrix norm on $\mathbb{R}^{N \times N}$. As a direct consequence of Gelfand's formula, we have

$$\lim_{n \rightarrow \infty} \|\mathcal{M}^n\|^{1/n} = \rho(\mathcal{M}),$$

and this bound is typically only asymptotically tight for most matrix norms. Thus, if $\rho(\mathcal{M}) < 1$, then the map $(H_{2,i})_{i \in [N]}$ is eventually contractive under any norm on $\prod_{i \in [N]} \mathcal{P}(X)$. With this observation, and using Lemma 4, we obtain the following meta contraction condition in the stationary case:

Theorem 1. If $\rho(\mathcal{M}) < 1$, then $\tau \mapsto (H_{2,i}(Q^{\tau,i}, \tau))_{i \in [N]}$ is eventually contractive over $\prod_{i \in [N]} \mathcal{P}(X)$ under any norm, and thus Algorithm 1 converges.

Proof. This follows from the discussion above. □

In general, finding a simple closed-form expression for $\rho(\mathcal{M})$ in terms of radicals is impossible (due to Abel-Ruffini theorem) as it amounts to finding a root of the characteristic equation of $\mathcal{M}$, which has degree $|N|$. However, since $\mathcal{M}$ is a nonnegative irreducible matrix, spectral radius $\rho(\mathcal{M})$ of $\mathcal{M}$ corresponds to an eigenvalue of $\mathcal{M}$ by the Perron-Frobenius theorem, which can be used to give a full characterization of $\rho(\mathcal{M}) < 1$.

Theorem 2. We have $\rho(\mathcal{M}) < 1$ if and only if

$$\sum_{i \in [N]} \left(K_i + \frac{\bar{L}_i K_i}{\rho_i(1 - \beta_i)} \right) \frac{1}{(1 - (\bar{K}_i - K_i))} < 1$$

Proof. Let h be a positive right eigenvector such that $\mathcal{M}h = \rho(\mathcal{M})h$. Then, using the equality of $\mathcal{M}h = \rho(\mathcal{M})h$ on the j 'th row, one can see that for all $j \in [N]$ we have

$$(\bar{K}_j - K_j)h_j + \underbrace{\sum_{i \in [N]} \left(K_i + \frac{\bar{L}_i K_i}{\rho_i(1 - \beta_i)} \right) h_i}_{=k} = \rho(\mathcal{M})h_j. \quad (5)$$

Note that, for all $j \in [N]$ we have $\rho(\mathcal{M}) > \bar{K}_j - K_j$ due to monotonicity of the spectral radius for nonnegative matrices. Thus, for all $j \in [N]$ we have

$$\frac{k}{\rho(\mathcal{M}) - (\bar{K}_j - K_j)} = h_j. \quad (6)$$

Equation (6) together with the definition of k on (5) imply that

$$k = \sum_{i \in [N]} \frac{\left(K_i + \frac{\bar{L}_i K_i}{\rho_i(1 - \beta_i)} \right)}{\rho(\mathcal{M}) - (\bar{K}_j - K_j)} k. \quad (7)$$

The equation (7) is feasible if and only if

$$\sum_{i \in [N]} \frac{\left(K_i + \frac{\bar{L}_i K_i}{\rho_i(1 - \beta_i)} \right)}{\rho(\mathcal{M}) - (\bar{K}_j - K_j)} = 1. \quad (8)$$

Note that the function

$$x \mapsto \sum_{i \in [N]} \frac{\left(K_i + \frac{\bar{L}_i K_i}{\rho_i(1 - \beta_i)} \right)}{x - (\bar{K}_j - K_j)} \quad (9)$$

is monotone decreasing on $(\max_j(\bar{K}_j - K_j), \infty)$; thus, there exists a unique $x > \max_j(\bar{K}_j - K_j)$ that satisfies (8), which is $\rho(\mathcal{M})$. Using the monotonicity of (9) together with (8), one can directly deduce that the function defined by (9) is less than 1 at $x = 1$ is equivalent to $\rho(\mathcal{M}) < 1$. $\square$

4 Variational Contraction Conditions for Finite-Horizon MFG

Let $\text{MFG}_T = (X, \mathcal{P}(A), (C_i + \rho_i \Omega_i)_{i \in [N]}, (P_i)_{i \in [N]}, N)$ be a finite-horizon regularized discrete-time MFG with horizon-length T . In this section, we will study the convergence behavior of a simple iterative algorithm to find a MFE for MFG_T under a given initial state-measure τ_0 .

4.1 Statements of Main Results

In this subsection, we construct a matrix whose spectral radius yields a contraction condition in the finite-horizon setting, in a manner similar to Section 3. Due to the multi-population setting and the non-stationarity of the solutions, the relevant contraction condition is determined by the spectral radius of an $N(T-1) \times N(T-1)$ dense matrix with no exploitable structure. We will derive a majorant for this contraction matrix that is independent of the horizon length and show that the spectral radius of these contraction matrices converges to that of the majorizing matrix, which will allow us to provide a horizon length independent contraction condition for MFG_T .

For a given $\tau = (\tau_t)_{t=0}^{T-1} \in \prod_{t=0}^{T-1} \left(\prod_{i=1}^N \mathcal{P}(X) \right)$, let τ_t denote the section of the flow τ at time step t , regarded as an element of $\prod_{i=1}^N \mathcal{P}(X)$. Similarly, for a given $t \in \{0, \dots, T-1\}$, let $\tau_{t,j}$ denote the component of τ_t corresponding to population j , regarded as an element of $\mathcal{P}(X)$.

Definition 3. Let $\tau = (\tau_j)_{j=1}^{T-1} \in \prod_{t=0}^{T-1} \left(\prod_{i=1}^N \mathcal{P}(X) \right)$. For each $i \in [N]$, we define a sequence of Q -functions $(Q_t^{\tau,i})_{t=0}^{T-1}$ by backward recursion, starting from $t = T - 1$:

$$Q_{T-1}^{\tau,i}(x, u) = C_i(x, u, \tau_{T-1}) + \rho_i \Omega_i(u),$$

and, for $t = 0, 1, \dots, T - 2$,

$$Q_t^{\tau,i}(x, u) = C_i(x, u, \tau_t) + \rho_i \Omega_i(u) + \beta_i \sum_{y \in X} Q_{\min, t+1}^{\tau,i}(y) P_i(y | x, u, \tau_t).$$

We denote by $\mathbf{Q}^{\tau,i} = (Q_t^{\tau,i})_{t=0}^{T-1}$ the resulting Q -function flow. Note that, by Lemma 3, we have $Q_t^{\tau,i} \in \mathcal{C}_i$ for all $i \in [N]$ and $t = 0, \dots, T - 1$.

Note that the flows $(\mathbf{Q}^{\tau,i})_{i \in [N]}$ are obtained under the discount factor $\beta < 1$. We will need the discount factor to analyze the long time behavior of finite-horizon MFE to identify a limiting system, which will be an infinite-horizon MFGs in our case. An iterative algorithm that finds a finite-horizon MFE starting from the initial state measure $\tau_0 \in \prod_{i \in [N]} \mathcal{P}(X)$ is the following:

Algorithm 2: Iteration Algorithm for Finding Finite-Horizon MFE

Let $\tau_0 = (\tau_{i,0})_{i \in [N]} \in \prod_{i \in [N]} \mathcal{P}(X)$ be an initial state-measure;
Let $\tau^0 := (\tau_0, \tau_1, \dots, \tau_{T-1}) \in \prod_{t=0}^{T-1} \left(\prod_{i \in [N]} \mathcal{P}(X) \right)$;
while $\tau^{n+1} \neq \tau^n$ **do**
 | $\tau^{n+1} = (\tau_{0,i}, H_{2,i}(Q_0^{\tau^n,i}, \tau_0^n), H_{2,i}(Q_1^{\tau^n,i}, \tau_1^n), \dots, H_{2,i}(Q_{T-1}^{\tau^n,i}, \tau_{T-1}^n))_{i \in [N]}$;
end
Define $\pi_{t,i}^* = \delta_{\arg\min_{u \in \mathcal{P}(A)} Q_t^{\tau^*,i}(\cdot, u)}(\cdot)$ for all $i \in [N]$ and $t = 0, \dots, T - 1$;
Let $\pi^* = (\pi_i^*)_{i \in [N]}$;
return finite-horizon mean-field equilibrium (π^*, τ^*)

The contraction map that we will consider that ensures converge of Algorithm 2 in this section will take a family of state-measure τ as an input, calculate $\mathbf{Q}^{\tau,i}$ for all $i \in [N]$, and then update τ by using the minimizers of $\mathbf{Q}^{\tau,i}$ as required for an MFE.

The main result of this section is the following contraction theorem, which provides a variational condition guaranteeing the contraction uniformly in the horizon length.

Theorem 3. Let $\bar{K}_\infty = \min_j (\bar{K}_j - K_j)$, and $a_i = \frac{\bar{L}_i K_i}{\rho_i}$. If

$$\inf_{\bar{K}_\infty < r < \beta_{\max}^{-1}} \sum_{i \in [N]} \frac{\frac{K_i + a_i}{r} + \frac{a_i \beta_i}{1 - \beta_i r}}{1 - \frac{\bar{K}_i - K_i}{r}} < 1, \quad (10)$$

then for all $T \in \mathbb{N}$, the map $\tau \mapsto \left((\tau_{0,i})_{i=1}^N, \left(H_{2,i}(Q_t^{\tau,i}, \tau_t) \right)_{i=1, t=1}^{N, T-1} \right)$ is eventually contractive over $\prod_{t=1}^{T-1} \left(\prod_{i=1}^N \mathcal{P}(X) \right) \times \prod_{i=1}^N \{\tau_{0,i}\}$ for any $(\tau_{0,i})_{i=1}^N \in \prod_{i=1}^N \mathcal{P}(X)$. In particular, when (10) holds, Algorithm 2 converges.

Proof. In the remainder of this section, we outline the main ideas of the proof. The full argument is developed in Sections 4.2, 4.3, and 4.4. $\square$

We emphasize that, although (10) provides an equivalent criterion for contractivity, it is not the contraction rate of the map

$$\tau \mapsto \left((\tau_{0,i})_{i=1}^N, \left(H_{2,i}(Q_t^{\tau,i}, \tau_t) \right)_{i=1, \dots, N, t=1, \dots, T-1} \right)$$

in $\prod_{t=1}^{T-1} \left(\prod_{i=1}^N \mathcal{P}(X) \right) \times \prod_{i=1}^N \{\tau_{0,i}\}$ for any $(\tau_{0,i}) \in \prod_{i=1}^N \mathcal{P}(X)$ under any norm. A particular case of Theorem 3 is the single-population setting, in which we recover a contraction condition equivalent to that of [6, Theorem 2].

Corollary 1. Suppose $N = 1$, i.e. we have a single population MFG. Then, we have contraction for the mean-field equilibrium operator if

$$\inf_{\bar{K}-K < r < \beta} \frac{\bar{K} + \frac{\bar{L}K}{\rho}}{r} + \frac{\bar{L}K\beta}{1-\beta r} < 1. \quad (11)$$

Proof. This follows directly from Theorem 3. $\square$

Remark 1. A condition that is equivalent to (11) is given in [6, Theorem 2]. The main advantage of the corollary above compared to [6, Theorem 2] is that, we will be able to obtain explicit convergence rates from finite-horizon to infinite-horizon MFE under the variational characterization under slightly weaker conditions than the ones used in [6].

The proof of Theorem 3 relies on finding a uniform upper bound on the spectral radius of a matrix as done in Section 3. The main motivation behind this result is that minimizers of the left-hand side of (10) allow us to construct a Lyapunov function that yields explicit convergence rates between finite-horizon and infinite-horizon non-stationary MFEs, which is the subject of the next section. These convergence rates also allow us to deduce a uniqueness condition for infinite-horizon non-stationary MFEs.

As in the stationary case, we begin our analysis by establishing a vector inequality for the finite-horizon setting that describes the worst-case behavior of the iterations obtained under the map $\tau \mapsto \left((\tau_{0,i})_{i=1}^N, \left(H_{2,i}(Q_t^{\tau,i}, \tau_t) \right)_{i=1, t=1}^{N, T-1} \right)$.

Lemma 5. For all $i \in [N]$, let $\bar{\mathcal{M}}_i, \mathcal{M}_i \in \mathbb{R}^{(T-1) \times (T-1)}$ be such that $\bar{\mathcal{M}}_i = T_i(\bar{K}_i)$ and $\mathcal{M}_i = T_i(K_i)$, where

$$T_i(x) = \begin{bmatrix} \frac{\bar{L}_i K_i}{\rho_i} \beta_i & \frac{\bar{L}_i K_i}{\rho_i} \beta_i^2 & \frac{\bar{L}_i K_i}{\rho_i} \beta_i^3 & \dots & \frac{\bar{L}_i K_i}{\rho_i} \beta_i^{T-1} \\ x + \frac{\bar{L}_i K_i}{\rho_i} & \frac{\bar{L}_i K_i}{\rho_i} \beta_i & \frac{\bar{L}_i K_i}{\rho_i} \beta_i^2 & \dots & \frac{\bar{L}_i K_i}{\rho_i} \beta_i^{T-2} \\ 0 & x + \frac{\bar{L}_i K_i}{\rho_i} & \frac{\bar{L}_i K_i}{\rho_i} \beta_i & \dots & \frac{\bar{L}_i K_i}{\rho_i} \beta_i^{T-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \frac{\bar{L}_i K_i}{\rho_i} \beta_i \end{bmatrix}.$$

By $S_{a,b}(\mathcal{M}_i, \mathcal{M}_j)$, denote an $a(T-1) \times b(T-1)$ dimensional block matrix with $\mathcal{M}_1$ on the diagonals and $\mathcal{M}_j$ on the off-diagonals. Let $E_{i,j}$ denote an $N \times N$ matrix with a 1 in the (i,j) -entry and zeros elsewhere. We define

$$\mathcal{S}_T = \sum_{i=1}^N E_{i,i} \otimes \mathcal{S}_{N_i, N_i}(\bar{\mathcal{M}}_i, \mathcal{M}_i) + \sum_{i,j: i \neq j}^N E_{i,j} \otimes \mathcal{S}_{N_i, N_j}(\mathcal{M}_i, \mathcal{M}_j)$$

Then, for any $\tau, \tilde{\tau} \in \prod_{t=1}^{T-1} \left(\prod_{i=1}^N \mathcal{P}(X) \right) \times \prod_{i=1}^N \{\tau_{0,i}\}$ we have

$$\begin{bmatrix} \|H_{2,1}(Q_0^{\tau,1}, \tau_0) - H_{2,1}(Q_0^{\tilde{\tau},1}, \tilde{\tau}_0)\|_1 \\ \|H_{2,1}(Q_1^{\tau,1}, \tau_1) - H_{2,1}(Q_1^{\tilde{\tau},1}, \tilde{\tau}_1)\|_1 \\ \vdots \\ \|H_{2,1}(Q_{T-2}^{\tau,1}, \tau_{T-2}) - H_{2,1}(Q_{T-2}^{\tilde{\tau},1}, \tilde{\tau}_{T-2})\|_1 \\ \|H_{2,2}(Q_0^{\tau,2}, \tau_0) - H_{2,2}(Q_0^{\tilde{\tau},2}, \tilde{\tau}_0)\|_1 \\ \|H_{2,2}(Q_1^{\tau,2}, \tau_1) - H_{2,2}(Q_1^{\tilde{\tau},2}, \tilde{\tau}_1)\|_1 \\ \vdots \\ \|H_{2,2}(Q_{T-2}^{\tau,2}, \tau_{T-2}) - H_{2,2}(Q_{T-2}^{\tilde{\tau},2}, \tilde{\tau}_{T-2})\|_1 \\ \vdots \\ \|H_{2,N}(Q_0^{\tau,N}, \tau_0) - H_{2,N}(Q_0^{\tilde{\tau},N}, \tilde{\tau}_0)\|_1 \\ \|H_{2,N}(Q_1^{\tau,N}, \tau_1) - H_{2,N}(Q_1^{\tilde{\tau},N}, \tilde{\tau}_1)\|_1 \\ \vdots \\ \|H_{2,N}(Q_{T-2}^{\tau,N}, \tau_{T-2}) - H_{2,N}(Q_{T-2}^{\tilde{\tau},N}, \tilde{\tau}_{T-2})\|_1 \end{bmatrix} \leq \mathcal{S}_T \begin{bmatrix} \|\tau_{1,1} - \tilde{\tau}_{1,1}\|_1 \\ \|\tau_{2,1} - \tilde{\tau}_{2,1}\|_1 \\ \vdots \\ \|\tau_{T-1,1} - \tilde{\tau}_{T-1,1}\|_1 \\ \|\tau_{1,2} - \tilde{\tau}_{1,2}\|_1 \\ \|\tau_{2,2} - \tilde{\tau}_{2,2}\|_1 \\ \vdots \\ \|\tau_{T-1,2} - \tilde{\tau}_{T-1,2}\|_1 \\ \vdots \\ \|\tau_{1,N} - \tilde{\tau}_{1,N}\|_1 \\ \|\tau_{2,N} - \tilde{\tau}_{2,N}\|_1 \\ \vdots \\ \|\tau_{T-1,N} - \tilde{\tau}_{T-1,N}\|_1 \end{bmatrix}.$$

Proof. Fix an arbitrary $i \in [N]$. For any $0 < t \leq T-2$, using Lemma 2, through recursive iterations, we obtain

$$\begin{aligned} & \|H_{2,i}(Q_t^{\tau,i}, \tau_t) - H_{2,i}(Q_t^{\tilde{\tau},i}, \tilde{\tau}_t)\|_1 \\ & \leq \frac{K_i}{\rho_i} \|Q_t^{\tau,i} - \tilde{Q}_t^{\tilde{\tau},i}\|_\infty + \bar{K}_i \|\tau_{t,i} - \tilde{\tau}_{t,i}\|_1 + K_i \sum_{j \neq t} \|\tau_{t,j} - \tilde{\tau}_{t,j}\|_1 \\ & \leq \frac{K_i \beta_i}{\rho_i} \|Q_{t+1}^{\tau,i} - \tilde{Q}_{t+1}^{\tilde{\tau},i}\|_\infty + \left(\bar{K}_i + \frac{\bar{L}_i K_i}{\rho_i} \right) \|\tau_{t,i} - \tilde{\tau}_{t,i}\|_1 + \left(K_i + \frac{\bar{L}_i K_i}{\rho_i} \right) \|\tau_{t,j} - \tilde{\tau}_{t,j}\|_1 \\ & \leq \left(\bar{K}_i + \frac{\bar{L}_i K_i}{\rho_i} \right) \|\tau_{t,i} - \tilde{\tau}_{t,i}\|_1 + \left(K_i + \frac{\bar{L}_i K_i}{\rho_i} \right) \|\tau_{t,j} - \tilde{\tau}_{t,j}\|_1 \\ & \quad + \frac{\bar{L}_i K_i}{\rho_i} \sum_{k=1}^{T-1-t} \beta^k \sum_j \|\tau_{t+k,j} - \tilde{\tau}_{t+k,j}\|_1. \end{aligned}$$

For $t = 0$, since $\tau_0 = \tilde{\tau}_0$, repeating the same argument above, we obtain that

$$\|H_{2,i}(Q_0^{\tau,i}, \tau_0) - H_{2,i}(Q_0^{\tilde{\tau},i}, \tilde{\tau}_0)\|_1 \leq \frac{\bar{L}_i K_i}{\rho_i} \sum_{k=1}^{T-1} \beta^k \sum_j \|\tau_{t+k,j} - \tilde{\tau}_{t+k,j}\|_1.$$

These two expressions give the desired matrix inequality. $\square$

Due to lack of sparsity of the matrix $\mathcal{S}_T$, numerical estimation of its spectral radius $\rho(\mathcal{S}_T)$ becomes more difficult as T increases. Moreover, the dimension of $\mathcal{S}_T$ also depends on the number of populations, which further complicates the problem. The asymptotic behavior of the ratios of eigenvector components corresponding to $\rho(\mathcal{S}_T)$ may also provide insight into finite-time error rates between finite-horizon and infinite-horizon MFEs; see [6, Theorem 4]. In general, we will not be able to provide a closed-form expression for $\rho(\mathcal{S}_T)$ or for $\lim_{T \rightarrow \infty} \rho(\mathcal{S}_T)$. Instead, we provide a contraction condition that can be computed efficiently numerically. In particular, to prove Theorem 3, we prove the following:

Theorem 4. *We have $\lim_{T \rightarrow \infty} \rho(\mathcal{S}_T) < 1$ if and only if*

$$\inf_{\bar{K}_\infty < r < \beta_{\max}^{-1}} \sum_{i \in [N]} \frac{\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i} \frac{1}{1 - \beta_i r}}{1 - \frac{\bar{K}_i - K_i}{r}} < 1. \quad (12)$$

Proof. A proof for this result will be established through sections 4.2, 4.3, and 4.4. $\square$

The difficult part of the proof of Theorem 4 is showing that $\lim_{T \rightarrow \infty} \rho(\mathcal{S}_T) < 1$ implies (12). Proving that (12) implies $\lim_{T \rightarrow \infty} \rho(\mathcal{S}_T) < 1$ will be argued through a Collatz-Wiedelant argument by choosing geometric test vectors. If $\lim_{T \rightarrow \infty} \rho(\mathcal{S}_T) < 1$, then part of the proof of Theorem 4 implies that $\bar{K}_\infty < r < \beta_{\max}^{-1}$. Thus, when (12) is satisfied with $r = 1$, we recover the contraction condition for stationary MFEs; see Theorem 2. In this sense, the contraction condition for finite-horizon MFEs is strictly weaker than that for stationary MFEs.

Let

$$r \in (\min_j (\bar{K}_j - K_j), \beta_{\max}^{-1}) \mapsto V(r) = \sum_{i \in [N]} \frac{\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i} \frac{1}{1 - \beta_i r}}{1 - \frac{\bar{K}_i - K_i}{r}}.$$

We call V a *variational contraction function* for the finite-horizon MFE. The value $\inf_r V(r)$ represents the *worst-case* behavior of the finite-horizon MFG that can be inferred from a majorization of $\mathcal{S}_T$. In the next section, when we derive finite-time error bounds between finite-horizon and infinite-horizon MFEs, the regime of the values of r such that $V(r) < 1$ will play a critical role. In general, we can separate the behavior of V into three regimes: $r < 1$, $r = 1$, and $r > 1$.

1. When $V(1) < 1$, there exists a majorization of $\mathcal{S}_T$ such that the ratios of consecutive entries of a positive Perron right eigenvector converge to the constant 1. Note that the case $V(1)$ corresponds to the contraction rate in Theorem 2; thus, the asymptotic stationarity of these eigenvectors implies that the system approaches consensus. In this case, we say that $\mathcal{S}_T$ is *asymptotically stationary*.
2. When $V(r) < 1$ for some $r > 1$, there exists a majorization of $\mathcal{S}_T$ such that the ratios of consecutive entries of a positive Perron right eigenvector converge to the constant r . In this case, we say that $\mathcal{S}_T$ is *stable*. This stability comes from the fact that the corresponding majorization has Perron eigenvectors that decay geometrically with ratio $1/r$. Stable cases are those in which we can recover a convergence rate from finite-horizon MFEs to infinite-horizon MFEs. Note that, by continuity of V , if $\mathcal{S}_T$ is asymptotically stationary, then it is also stable.
3. When $V(r) < 1$ only for $r < 1$, there exists a majorization of $\mathcal{S}_T$ such that the ratios of consecutive entries of a positive Perron right eigenvector converge to the constant r . In this case, we say that $\mathcal{S}_T$ is *unstable*.

The results in [6, Theorems 2 and 4] are equivalent to the worst-case quantity $\inf_r V(r)$ in the single-population setting. For instance, as we show in the next section, when the stationary MFG is eventually contractive, i.e., when $V(1) < 1$, we can still recover a convergence rate between finite-horizon and infinite-horizon MFEs using the variational contraction function V . In this sense, the result above generalizes [6, Theorem 2].

Our proof strategy for Theorem 4 amounts to showing that appropriate majorizations of $\mathcal{S}_T$ with the stated characteristics exist in each of the three regimes described above. These characteristics are identified by studying a limiting eigenvector equation derived from the finite-horizon Perron right eigenvectors of $\mathcal{S}_T$. Note that for each $T \in \mathbb{N}$, the Perron right eigenvector of $\mathcal{S}_T$ satisfies

$$\rho(\mathcal{S}_T) u_i(1) = \frac{\bar{L}_i K_i}{\rho_i} \sum_{j \in [N]} \sum_{k=1}^{T-1} \beta_i^k u_j(k), \quad i \in [N],$$

and

$$\rho(\mathcal{S}_T) u_i(k) = (\bar{K}_i - K_i) u_i(k-1) + \sum_{j \in [N]} K_i u_j(k-1) + \frac{\bar{L}_i K_i}{\rho_i} \sum_{j \in [N]} \sum_{t=k-1}^{T-1} \beta_i^{t-k+1} u_j(t),$$

for $i \in [N]$ and $2 \leq k \leq T-1$. In what follows, we let $\lambda_T := \rho(\mathcal{S}_T)$.

Let $\lambda_\infty = \lim_{T \rightarrow \infty} \lambda_T$, which exists since the sequence $(\lambda_T)_{T \geq 0}$ is monotone. On the space of real sequences $(\mathbb{R}^{\mathbb{N}})^N$, for $u = (u_i(k))_{k \in \mathbb{N}, i \in [N]}$ and any $\lambda > 0$, we define the *limiting eigenvector operator* $\mathcal{T}_\lambda$ by taking $T \rightarrow \infty$ in the relations above:

$$(\mathcal{T}_\lambda(u))_i(1) = \frac{\bar{L}_i K_i}{\rho_i \lambda} \sum_{k \geq 1} \beta_i^k \sum_{j \in [N]} u_j(k), \quad i \in [N] \quad (13)$$

and for $k \geq 2$

$$(\mathcal{T}_\lambda(u))_i(k) = \frac{(\bar{K}_i - K_i)}{\lambda} u_i(k-1) + \frac{K_i}{\lambda} \sum_{j \in [N]} u_j(k-1) + \frac{\bar{L}_i K_i}{\rho_i \lambda} \sum_{m \geq 0} \beta_i^m \sum_{j \in [N]} u_j(k+m-1), \quad i \in [N]. \quad (14)$$

Both in the finite-horizon and infinite-horizon cases, the conditions satisfied under $k = 1$ (i.e. (13)) will be referred to as *boundary conditions*.

Informally, we will show that if $\mathcal{T}_{\lambda_\infty}$ has a fixed point $u \in (\mathbb{R}^{\mathbb{N}})^N$ with strictly positive components, then for all $i \in [N]$ we have

$$\lim_{k \rightarrow \infty} \frac{u_i(k+1)}{u_i(k)} = \text{constant},$$

and this constant is the same for all $i \in [N]$. This limit corresponds to a parameter $r^* \in (\max_j (\bar{K}_j - K_j), \beta_{\max}^{-1})$ satisfying $\inf_r V(r) = V(r^*)$. The value(s) of r^* that attain $V(r^*)$ will be crucial for deriving finite-time error bounds between finite-horizon and infinite-horizon MFEs, which is the subject of the next section. To prove Theorem 4, we will need the following necessary quantitative criterion obtain for the existence of a positive fixed-point of $\mathcal{T}_\lambda$, which also might be of independent interest:

Theorem 5. *Let $\max_j \lambda / (\bar{K}_j - K_j) > 1$. If the operator $\mathcal{T}_\lambda$ has a positive fixed point then*

$$1 = \sum_{i \in [N]} \frac{K_i z_0 + \frac{\bar{L}_i K_i}{\rho_i} \frac{z_0}{z_0 - \beta_i}}{\lambda - (\bar{K}_i - K_i) z_0} \quad (15)$$

for some $z_0 \in (\beta_{\max}, \max_j \lambda / (\bar{K}_j - K_j))$.

Proof. A proof for this will be established through sections 4.2, 4.3, and 4.4. $\square$

When $N = 1$, the theorem above provides a criterion for the existence of a positive fixed-point of a one-sided infinite Toeplitz operator. In general, infinite dimensional Toeplitz operators are known to be non-compact; thus, their spectrum cannot be approximated by finite truncations. A particular implication of [6, Proposition 3] in the single population case is that one does not have $\lim_{T \rightarrow \infty} \rho(\mathcal{S}_T) = \rho(\mathcal{T}_{\lambda_\infty})$ when $\mathcal{T}_{\lambda_\infty}$ is defined on the space of bounded sequences $(\ell_\infty, \|\cdot\|_\infty)$, where $\|\cdot\|_\infty$ is the uniform norm. Part of the proof of Theorem 5 relies on identification of a weighted space where we have $\lim_{T \rightarrow \infty} \rho(\mathcal{S}_T) = \rho(\mathcal{T}_{\lambda_\infty})$.

For our purposes, the proof of Theorem 4 has three main components:

1. The operator $\mathcal{T}_{\lambda_\infty}$ has a positive fixed point in $(\mathbb{R}^{\mathbb{N}})^N$ (Proposition 2).
2. Let $\lambda > 0$. If $\mathcal{T}_\lambda$ has a positive fixed point $u \in (\mathbb{R}^{\mathbb{N}})^N$, then for its fixed point $(u_i(k))_{i,k}$, $\lim_{k \rightarrow \infty} \frac{u_i(k+1)}{u_i(k)}$ exists and equals a constant independent of $i \in [N]$ (Lemma 18).
3. If $\mathcal{T}_\lambda$ has a positive fixed point and $\lambda > \max_j (\bar{K}_j - K_j)$, then $\lambda = \rho(B(r))$ for some r (Lemmas 19 and 20), where $\rho(B(r))$ denotes the spectral radius of the matrix $B(r)$ defined by

$$B(r) := D(r) + \mathbf{1} v(r)^\top.$$

Here, $\mathbf{1}$ is the vector of ones and

$$v_j(r) = \frac{K_j}{r} + \frac{\bar{L}_j K_j}{\rho_j (1 - \beta_j r)}, \quad D(r) := \text{diag} \left(\frac{\bar{K}_1 - K_1}{r}, \frac{\bar{K}_2 - K_2}{r}, \dots, \frac{\bar{K}_N - K_N}{r} \right),$$

$$v(r) := (v_1(r), \dots, v_N(r))^\top.$$

The spectral radius of $B(r)$ provides a majorization of the spectral radius of $\mathcal{S}_T$ for all T . Using multiple copies of $B(r)$, we can also construct a matrix that directly majorizes $\mathcal{S}_T$; this will be done in the next section when deriving convergence rates from finite-horizon to infinite-horizon MFEs. The constraint (15) then will be shown to hold for $\rho(B(r))$.

The first item will be proved using the dominated convergence theorem by showing the uniform integrability of Perron eigenvectors of $\mathcal{S}_T$ over the counting measure. To show this, we will heavily rely on the geometric decaying tails of the matrices $\mathcal{S}_T$ and work with normalized Perron eigenvectors (Proposition 2), which establishes an asymptotic regularity property for these Perron eigenvectors.

The second item will be proved by mapping any positive fixed point of $\mathcal{T}_\lambda$ to the complex plane using generating functions, inspired by methods from analytic combinatorics. As in the argument establishing regularity for the Perron right eigenvectors of $\mathcal{S}_T$, we will show that these generating functions admit a sufficiently large radius of convergence around the origin $z = 0$ (Lemmas 11 and 14). We then use basic tools from singularity analysis in complex analysis to conclude the result (Lemma 18).

The first two items above then will be sufficient to provide a sharp upper bound on $\rho(\mathcal{S}_T)$ that is independent of T . The last item (combined with the second item) is needed to obtain the characterization in Theorem 4 using Theorem 5. An important feature of this step is that it avoids computing $\inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r))$ as an intermediate step. Nevertheless, if one is interested in the asymptotics of $(\rho(\mathcal{S}_T))_T$, a useful application of the third item is the following limit result for the asymptotic behavior of $(\rho(\mathcal{S}_T))_T$.

Proposition 1. *We have*

$$\lim_{T \rightarrow \infty} \rho(\mathcal{S}_T) = \inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r)).$$

Proof. We postpone the proof to Section 4.4 □

The advantage of the calculation above, compared with computing $\rho(\mathcal{S}_T)$ directly, is that the matrix $B(r)$ is independent of the horizon length T . It can also be shown that $B(r)$ can be reduced to depend only on distinct population types, i.e., we can collapse identical populations into a single population.

4.2 Step 1: Upper Bound $\lambda_\infty \leq \inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r))$

In this subsection, we show that the bound

$$\lambda_\infty \leq \inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r))$$

holds, and that for any $0 < r < \beta_{\max}^{-1}$,

$$1 = \sum_{i \in [N]} \frac{\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i} \frac{1}{1 - \beta_i r}}{\rho(B(r)) - \frac{\bar{K}_i - K_i}{r}}.$$

To obtain the upper bound $\lambda_\infty \leq \inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r))$, we use the Collatz–Wielandt inequalities with geometric test vectors. We will later show that this geometric ansatz yields a sharp upper bound on $\rho(\mathcal{S}_T)$ as $T \rightarrow \infty$, as outlined in the previous subsection. Moreover, the majorization of λ_∞ by $\rho(B(r))$ will also allow us to derive convergence rates from finite-horizon to infinite-horizon MFEs.

We recall the Collatz–Wielandt bound for completeness.

Lemma 6 (Collatz-Wielandt bound). *For any nonnegative square matrix A and any positive vector $x > 0$ (of appropriate dimension), it holds that*

$$\rho(A) \leq \max_i \frac{(Ax)_i}{x_i}.$$

Proof. This follows from [14, Theorem 8.3.2]. $\square$

Lemma 7. *For every $T \geq 3$, and $r \in (0, \beta_{\max}^{-1})$, we have*

$$\lambda_T \leq \inf_{r \in (0, \beta_{\max}^{-1})} \rho(B(r)),$$

where $\rho(B(r))$ is the spectral radius of the matrix $B(r)$.

Proof. We will use the Collatz-Wielandt bound (Lemma 6) for a vector x where each component is a power of some $r \in \mathbb{R}$. For this purpose, we start with fixing r with $0 < r < (\beta_{\max})^{-1}$, and define $x = x(r) \in \mathbb{R}_{>0}^{T-1}$ by

$$x_i := r^{i-1} \quad (i = 1, \dots, T-1).$$

For $c = (c_1, \dots, c_N) \in \mathbb{R}_{>0}^N$, we define $\bar{x} = \bar{x}(r) = (c_1 x(r), c_2 x(r), \dots, c_N x(r))$. We also define $C = \sum_{i \in [N]} c_i$, and $g = \sum_{i \in [N]} c_i x(r) = Cx(r)$.

For each $i \in [T-1]$, and $i \geq 2$, for any $m \in [N-1]$ we have

$$(\mathcal{S}_T g)_{m(T-1)+i} = (\bar{K}_m - K_m) c_m x_{i-1} + C \left(K_m + \frac{\bar{L}_m K_m}{\rho_m} \right) x_{i-1} + C \sum_{j=i}^{T-1} \beta_m^{j-i+1} x_j;$$

thus,

$$\begin{aligned} \frac{(\mathcal{S}_T g)_{m(T-1)+i}}{c_m x_i} &= \frac{\bar{K}_m - K_m}{r} + \frac{C \left(K_m + \frac{\bar{L}_m K_m}{\rho_m} \right)}{c_m r} + \frac{C \bar{L}_m K_m \sum_{j=i}^{T-1} \beta_m^{j-i+1} r^{j-i}}{\rho_m c_m} \\ &\leq \frac{\bar{K}_m - K_m}{r} + \frac{C \left(K_m + \frac{\bar{L}_m K_m}{\rho_m} \right)}{c_m r} + \frac{C \bar{L}_m K_m \sum_{j=1}^{\infty} \beta_m^j r^j}{\rho_m c_m} \\ &\leq \frac{\bar{K}_m - K_m}{r} + \frac{C \left(K_m + \frac{\bar{L}_m K_m}{\rho_m} \right)}{c_m r} + \frac{C \bar{L}_m K_m \beta_m}{\rho_m c_m (1 - \beta_m r)} \\ &= \frac{(B(r)c^\top)_m}{c_m}. \end{aligned}$$

This implies that

$$\inf_{c \in \mathbb{R}_{>0}^N} \max_i \frac{(\mathcal{S}_T g)_{m(T-1)+i}}{c_m x_i} \leq \inf_{c \in \mathbb{R}_{>0}^N} \frac{(B(r)c^\top)_m}{c_m} = \rho(B(r)),$$

where the last equality is due to the Perron-Frobenius theorem.

Using the Collatz Wiedelant bound (Lemma 6), for all $r \in (0, \beta_{\max}^{-1})$ it holds

$$\lambda_T \leq \rho(B(r)) \implies \lambda_T \leq \inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r)).$$

To pass to the limit as $T \rightarrow \infty$, note that $\lambda_T \leq \lambda_{T+1}$ for all T , since we have “ $\mathcal{S}_T \leq \mathcal{S}_{T+1}$ ” for all T , which can be proved by embedding $\mathcal{S}_T$ into top left corner of $\mathcal{S}_{T+1}$. Thus, $\lim_{T \rightarrow \infty} \lambda_T =: \lambda_\infty$ exists due to monotonicity. Thus, we must have

$$\lambda_\infty \leq \inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r)),$$

as desired. $\square$

Lemma 7 also serves as the first half of the proof of Proposition 1 as we have shown that $\lim_{T \rightarrow \infty} \rho(\mathcal{S}_T) \leq \inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r))$.

Next, we establish a relation between $\rho(B(r))$ and the parameter r . This will later be used to relate the values of $\lambda > 0$ for which $\mathcal{T}_\lambda$ admits a positive fixed point to corresponding values of $\rho(B(r))$.

Lemma 8. *For any $0 < r < \beta_{\max}^{-1}$, we have*

$$1 = \sum_{i \in [N]} \frac{\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i} \frac{1}{1 - \beta_i r}}{\rho(B(r)) - \frac{\bar{K}_i - K_i}{r}}.$$

Furthermore, it holds that

$$\beta_{\max}^{-1} > r > \min_{j \in [N]} \frac{\bar{K}_j - K_j}{\rho(B(r))}.$$

Proof. The proof closely follows that of Theorem 2. Since $B(r)$ is a nonnegative irreducible matrix, by the Perron-Frobenius theorem, it holds that there exists a nonnegative vector $\bar{c} = (c_1, c_2, \dots, c_N)$ such that $B(r)\bar{c} = \rho(B(r))\bar{c}$. Then, for any $m \in [N]$, we have

$$(\bar{K}_m - K_m) \frac{c_m}{r} + k = \rho(B(r))c_m \implies k = \left(\rho(B(r)) - (\bar{K}_m - K_m) \frac{1}{r} \right) c_m,$$

where

$$k = \sum_{i \in [N]} \left(\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i(1 - \beta_i r)} \right) c_i. \quad (16)$$

Thus,

$$k = \sum_{i \in [N]} \left(\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i(1 - \beta_i r)} \right) c_i = \sum_{i \in [N]} \frac{\left(\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i(1 - \beta_i r)} \right)}{\rho(B(r)) - \frac{\bar{K}_i - K_i}{r}} k. \quad (17)$$

Note that $\rho(B(r)) > \frac{\bar{K}_i - K_i}{r}$ for all $i \in [N]$ due to monotonicity of spectral radius with respect to diagonal terms; so, it holds that

$$\frac{\left(\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i(1 - \beta_i r)} \right)}{\rho(B(r)) - \frac{\bar{K}_i - K_i}{r}} > 0.$$

Thus, as desired, (17) holds if and only if

$$\sum_{i \in [N]} \frac{\left(\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i(1 - \beta_i r)} \right)}{\rho(B(r)) - \frac{\bar{K}_i - K_i}{r}} = 1.$$

□

4.3 Step 2: Existence of a positive fixed point of $\mathcal{T}_{\lambda_\infty}$

In this subsection, we will show that for λ_∞ , there exists a positive fixed point of the operator $\mathcal{T}_{\lambda_\infty}$. To show this, as discussed before, we will use the right Perron eigenvectors of $\mathcal{S}_T$ under a particular normalization. Using the structure of the matrices $\mathcal{S}_T$, we will show that these Perron eigenvectors admit sufficient regularity conditions to use the dominated convergence theorem.

First, we show that, under a suitable normalization, the Perron right eigenvectors of $\mathcal{S}_T$ admit a convergent subsequence as $T \rightarrow \infty$. We choose a normalization that fixes the first coordinate of the population 1. Accordingly, we relate the first coordinates of the other populations to that of population 1 using the boundary conditions of the operator $\mathcal{T}_{\lambda_\infty}$.

Lemma 9 (Finite-horizon geometric lower bounds). *Fix $T \geq 2$ and let $\lambda_T > 0$ be the Perron eigenvalue of $\mathcal{S}_T$. Let $u_i^{(T)}(k) > 0$ denote a positive right Perron eigenvectors of $\mathcal{S}_T$, and define the aggregate $S^{(T)}(k) := \sum_{j=1}^N u_j^{(T)}(k)$ for $k = 1, \dots, T-1$. Write $\Delta_i := \bar{K}_i - K_i > 0$. We normalize eigenvectors by setting $u_1^{(T)}(1) = 1$ for all $T \in \mathbb{N}$.*

Then, for every $i \in [N]$ and every $k = 2, \dots, T-1$,

$$u_i^{(T)}(k) \geq \frac{\Delta_i}{\lambda_T} u_i^{(T)}(k-1). \quad (18)$$

Consequently, for all $k = 1, \dots, T-1$,

$$u_i^{(T)}(k) \geq \left(\frac{\Delta_i}{\lambda_\infty} \right)^{k-1} \frac{\bar{L}K_i}{\rho_i \lambda_\infty}. \quad (19)$$

Moreover, we have the upper bound

$$\left(\frac{\rho_i \lambda_\infty}{\bar{L}_i K_i \beta_i^2} \right)^{k-1} \frac{\rho_1 \lambda_\infty}{\bar{L}_1 K_1 \beta_1} \geq u_i^{(T)}(k).$$

Proof. First, we would like to set uniform upper and lower bounds for the terms $u_i^{(T)}(1)$. Since we set $u_1^{(T)}(1) = 1$, note that the boundary conditions give

$$u_i^{(T)}(1) \geq \frac{\bar{L}_i K_i}{\rho_i \lambda_T} \beta_i u_1^{(T)}(1) \geq \frac{\bar{L}_i K_i}{\rho_i \lambda_\infty} \beta_i u_1^{(T)}(1) = \frac{\bar{L}_i K_i}{\rho_i \lambda_\infty} \beta_i.$$

Similarly, we must also have that

$$\lambda_T u_1^{(T)}(1) \geq \frac{\bar{L}_1 K_1}{\rho_1} \beta_1 u_i^{(T)}(1) \implies \frac{\rho_1 \lambda_\infty}{\bar{L}_1 K_1 \beta_1} \geq u_i^{(T)}(1).$$

For $k = 2, \dots, T-1$, the finite-horizon reduced Perron equations have the form

$$\lambda_T u_i^{(T)}(k) = \left(K_i + \frac{\bar{L}_i K_i}{\rho_i} \right) S^{(T)}(k-1) + (\bar{K}_i - K_i) u_i^{(T)}(k-1) + \frac{\bar{L}_i K_i}{\rho_i} \sum_{m=1}^{T-k} \beta_i^m S^{(T)}(k+m-1), \quad (20)$$

where all coefficients and all terms on the right-hand side are nonnegative. Dropping the nonnegative terms $\left(K_i + \frac{\bar{L}_i K_i}{\rho_i} \right) S^{(T)}(k-1)$ and the future-tail sum yields

$$\lambda_T u_i^{(T)}(k) \geq \Delta_i u_i^{(T)}(k-1),$$

which proves (18). Iterating from 2 up to k gives

$$u_i^{(T)}(k) \geq \left(\frac{\Delta_i}{\lambda_T} \right)^{k-1} u_i^{(T)}(1) \geq \left(\frac{\Delta_i}{\lambda_\infty} \right)^{k-1} \frac{\bar{L}_i K_i}{\rho_i \lambda_\infty},$$

i.e., (19).

For the upper bound, using the boundary case of the eigenvector equation, i.e. when $k = 1$, we obtain

$$\lambda_T u_i^{(T)}(1) \geq \frac{\bar{L}_i K_i}{\rho_i} \beta_i^2 u_i^{(T)}(2).$$

For $k \geq 2$, we now have that

$$\lambda_T u_i^{(T)}(k) \geq \frac{\bar{L}_i K_i}{\rho_i} \beta_i^2 u_i^{(T)}(k+1).$$

Thus, we obtain the desired uniform upper bound

$$\left(\frac{\rho_i \lambda_\infty}{\bar{L}_i K_i \beta_i^2} \right)^{k-1} \frac{\rho_1 \lambda_\infty}{\bar{L}_1 K_1 \beta_1} \geq \left(\frac{\rho_i \lambda_T}{\bar{L}_i K_i \beta_i^2} \right)^{k-1} u_i^{(T)}(1) \geq u_i^{(T)}(k).$$

□

Let $(u_i^{(T)}(k))_{i,k}$ be a positive Perron right eigenvector of $\mathcal{S}_T$ normalized so that $u_1^{(T)}(1) = 1$. As a consequence of Lemma 9, and using a diagonal argument, we can extract a subsequence $(T_n)_{n \in \mathbb{N}} \subset \mathbb{N}$ such that for all $i \in [N]$ and $k \in \mathbb{N}$,

$$\lim_{n \rightarrow \infty} u_i^{(T_n)}(k) =: u_i(k) < \infty$$

for some family of sequences $(u_i(k))_{i,k}$. In particular, $u_i(k) > 0$ for all $k \geq 1$ and $i \in [N]$. To show that $(u_i(k))_{i,k}$ is a positive fixed point of the limiting eigenvector equation $\mathcal{T}_{\lambda_\infty}$, it suffices to justify the interchange of limit and summation:

$$\lim_{n \rightarrow \infty} \sum_{k=1}^{T_n-1} \beta_i^k \sum_{i \in [N]} u_i^{(T_n)}(k) = \sum_{k \geq 1} \beta_i^k \sum_{i \in [N]} u_i(k).$$

In the next lemma, we justify this exchange of limit and summation by showing that finite-horizon normalized right Perron eigenvectors are uniformly integrable under the geometric weights β_i^k ; this will be the key step in obtaining a positive fixed point for $\mathcal{T}_{\lambda_\infty}$.

Lemma 10. *Fix $i \in [N]$ with $a_i > 0$ and $\beta_i \in (0, 1)$. For each $T \geq 3$, let $u_i^{(T)}(k) > 0$ ($k = 1, \dots, T-1$) be the Perron right eigenvector of $\mathcal{S}_T$ corresponding to group i , normalized by*

$$u_1^{(T)}(1) = 1,$$

and extend it by setting $u_i^{(T)}(k) = 0$ for all $k \geq T$. Assume that $\lambda_T \rightarrow \lambda_\infty \in (0, \infty)$ and that, for each fixed $k \geq 1$, the limits

$$u_i^{(T)}(k) \longrightarrow u_i(k) \quad \text{as } T \rightarrow \infty$$

exist. Then,

$$\lim_{T \rightarrow \infty} \sum_{k \geq 1} \beta_i^k u_i^{(T)}(k) = \sum_{k \geq 1} \beta_i^k u_i(k).$$

Proof. Define the truncated tail sums

$$S_{k,T} := \sum_{m=k}^{T-1} \beta_i^{m-1} u_i^{(T)}(m), \quad k = 1, \dots, T-1.$$

From the finite-horizon tail inequality (obtained by dropping nonnegative terms in the k th eigenvector equation),

$$u_i^{(T)}(k) \geq \frac{\bar{L}_i K_i}{\rho_i \lambda_T} \sum_{j=1}^{T-1-k} \beta_i^j u_i^{(T)}(k+j), \quad k = 1, \dots, T-2,$$

let $c_{i,T} := (\rho_i \lambda_T) / (\bar{L}_i K_i)$. Multiplying by β_i^{k-1} yields

$$S_{k+1,T} \leq c_{i,T} \beta_i^{k-1} u_i^{(T)}(k), \quad k = 1, \dots, T-2.$$

Since $S_{k,T} = \beta_i^{k-1} u_i^{(T)}(k) + S_{k+1,T}$, we have $\beta_i^{k-1} u_i^{(T)}(k) = S_{k,T} - S_{k+1,T}$ and therefore

$$S_{k+1,T} \leq c_{i,T} (S_{k,T} - S_{k+1,T}) \implies S_{k+1,T} \leq \frac{c_{i,T}}{1 + c_{i,T}} S_{k,T}.$$

Iterating from 2 to k gives

$$S_{k,T} \leq S_{2,T} \left(\frac{c_{i,T}}{1 + c_{i,T}} \right)^{k-2}, \quad k = 2, \dots, T-1.$$

Because $\beta_i^{k-1} u_i^{(T)}(k) \leq S_{k,T}$, multiplying by β_i implies the key estimate:

$$\beta_i^k u_i^{(T)}(k) \leq S_{2,T} \left(\frac{c_{i,T}}{1 + c_{i,T}} \right)^{k-2}, \quad k = 2, \dots, T-1. \quad (21)$$

Next, under the normalization $u_1^{(T)}(1)$ for all $i \in [N]$ and $T \in \mathbb{N}$, apply the tail inequality at $k = 1$:

$$\frac{\rho_1 \lambda_\infty}{\bar{L}_1 K_1 \beta_1} u_i^{(T)}(1) \geq u_i^{(T)}(1) \geq \frac{\bar{L}_i K_i \beta_i}{\rho_i \lambda_T} \sum_{m=2}^{T-1} \beta_i^{m-1} u_i^{(T)}(m) = \frac{\bar{L}_i K_i \beta_i}{\rho_i \lambda_T} S_{2,T},$$

so $S_{2,T} \leq (\rho_i \lambda_T) / (\bar{L}_i K_i \beta_i) = c_{i,T}$. Substituting into (21) yields

$$\beta_i^k u_i^{(T)}(k) \leq c_{i,T} \frac{\rho_1 \lambda_\infty}{\bar{L}_1 K_1 \beta_1} \left(\frac{c_{i,T}}{1 + c_{i,T}} \right)^{k-2}, \quad k \geq 2.$$

Since $\lambda_T \rightarrow \lambda_\infty$, and $(\lambda_T)_T$ is increasing, for all T it holds that $\lambda_T \leq 2\lambda_\infty$, hence

$$c_{i,T} = \frac{\rho_i \lambda_T}{\bar{L}_i K_i} \leq \frac{\rho_i 2\lambda_\infty}{\bar{L}_i K_i} =: C_i, \quad \frac{c_{i,T}}{1 + c_{i,T}} = \frac{\lambda_T}{\lambda_T + \frac{\bar{L}_i K_i}{\rho_i}} \leq \frac{2\lambda_\infty}{2\lambda_\infty + \frac{\bar{L}_i K_i}{\rho_i}} =: q_i < 1.$$

Therefore, for all $T \geq 0$ and all $k \geq 2$,

$$0 \leq \beta_i^k u_i^{(T)}(k) \leq C_i \frac{\rho_1 \lambda_\infty}{\bar{L}_1 K_1 \beta_1} q_i^{k-2},$$

which gives T -independent uniform bound on the Perron eigenvectors. Since $q_i < 1$, and for all C_i there exists some $K > 0$ such that $C_i \leq K$, the majorant $C_i \frac{\lambda_\infty}{\bar{L}_1 \beta_1} q_i^{k-2}$ is summable over $k \geq 2$, and by assumption $\beta_i^k u_i^{(T)}(k) \rightarrow \beta_i^k u_i(k)$ pointwise for each fixed k .

The upper bounds we have established above imply that the family $(\beta_i^k u_i^{(T)}(k))_{i,k}$ is uniformly integrable under the counting measure. Hence, the dominated convergence theorem applied to the counting measure gives

$$\lim_{T \rightarrow \infty} \sum_{k \geq 2} \beta_i^k u_i^{(T)}(k) = \sum_{k \geq 2} \beta_i^k u_i(k).$$

Finally, adding the $k = 1$ terms: $\beta_i u_i^{(T)}(1) \rightarrow \beta_i u_i(1)$, yields the limit

$$\lim_{T \rightarrow \infty} \sum_{k \geq 1} \beta_i^k u_i^{(T)}(k) = \sum_{k \geq 1} \beta_i^k u_i(k),$$

as claimed. $\square$

In the next proposition, we state formally the existence of a positive fixed point of $\mathcal{T}_{\lambda_\infty}$, which we will reference later.

Proposition 2. *Let $\mathbf{u}^T = (u_i^{(T)}(k))_{i,k}$ be a positive Perron right eigenvector of $\mathcal{S}_T$, and normalize it by setting $u_1^{(T)}(1) = 1$. Then any accumulation point of the sequence $\mathbf{u}^T$ in the space of sequences $\mathbb{R}^{\mathbb{N}}$ (endowed with the product topology) as $T \rightarrow \infty$ is a fixed point of $\mathcal{T}_{\lambda_\infty}$. In particular, $\mathcal{T}_{\lambda_\infty}$ has a positive fixed point.*

Proof. This follows after writing the eigenvector equations for $(u_i^{(T)}(k))_{i,k}$ and applying Lemma 10 to exchange the limit and the summation. Since each section of $(u_i^{(T)}(k))_{i,k}$ is positive and bounded away from 0 by Lemma 9, it follows that the accumulation points of $(u_i^{(T)}(k))_{i,k}$ are also positive sequences, and hence, $\mathcal{T}_{\lambda_\infty}$ admits a positive fixed point. Note that the limits of the sections are also finite by Lemma 9. $\square$

4.4 Step 3: Limiting Behavior of the Ratio of Consecutive Terms of Perron Eigenvectors

In this subsection, we present the final step in the proof of Theorem 3 by filling in the missing details from the previous section. In particular, we prove that the fixed points of $\mathcal{T}_\lambda$ satisfying the limiting eigenvector equation (corresponding to $T = \infty$) imply that the ratios of consecutive components converge to the same limit. To this end, we use ideas from analytic combinatorics and translate the limiting solutions to the limiting eigenvector equation to the complex plane via generating functions associated with the eigenvectors and the “bulk” term. We then apply singularity analysis to show that the ratios of consecutive eigenvector components converge to a common constant.

The main difficulty in this step is to establish an appropriate radius of convergence for these generating functions. An initial estimate for the radius of convergence will be obtained by showing that positive fixed points of $\mathcal{T}_\lambda$ satisfy sufficient regularity conditions, as in Lemma 10. We then obtain the desired radius of convergence by constructing suitable analytic extensions of these generating functions after a series of algebraic manipulations.

To this end, we decompose the equations induced by $\mathcal{T}_\lambda$ into two parts: the eigenvector component $u_i(k)$ and the “bulk” term $s(k)$, defined as the sum of eigenvector components at time k . We first perform singularity analysis on the generating function of s . After deriving a relationship between s and u_i , we extend the singularity analysis to the generating function of u_i .

Let $(u_i(k))_{k \in \mathbb{N}}$ denote the limiting eigenvectors for $i \in [N]$, i.e., a positive fixed point of $\mathcal{T}_{\lambda_\infty}$, whose existence is guaranteed by Proposition 2. We assume that $u_1(1) = 1$. For most of this section, we will not rely on any special property of the constant λ_∞ ; thus, one may instead consider an arbitrary $\lambda > 0$ and assume that $(u_i(k))_{k \in \mathbb{N}}$ is a positive fixed point of $\mathcal{T}_\lambda$. The normalization $u_1(1) = 1$ is imposed to ensure that each component of the fixed point is bounded away from zero.

We define the generating function on $\mathbb{C}$ corresponding to the sequence $(u_i(k))_{k \in \mathbb{N}}$ by

$$U_i(z) := \sum_{k \in \mathbb{N}} u_i(k) z^k.$$

Let $s(k) = \sum_{i \in [N]} u_i(k)$. Similarly, we define the generating function corresponding to the bulk eigenvector term $s(k)$, associated with $(u_i(k))_{k \in \mathbb{N}, i \in [N]}$, as

$$S(z) := \sum_{k \in \mathbb{N}} s(k) z^k = \sum_{i \in [N]} U_i(z). \quad (22)$$

With these generating functions, our aim is to express $U_i(z)$ in terms of $S(z)$. We will study the behavior of the coefficients $u_i(k)$ via the series representation of U_i , for which we require a suitable radius of convergence.

Lemma 11. *For each $i \in [N]$, let R_i denote the radius of convergence of U_i about $z = 0$. Then $R_i > \beta_{\max}$ for all $i \in [N]$. Moreover, $R_i = R_j$ for all $i, j \in [N]$.*

Proof. The proof closely follows that of Lemma 10. The key step is establishing the strict inequality $R_i > \beta_{\max}$. We first show that for each $i \in [N]$, the radius of convergence satisfies $R_i > \beta_i$. Using the eigenvector equalities, note that for all $k \in \mathbb{N}$, using the boundary conditions that correspond to the population i , we must have

$$u_i(k) \geq \frac{\bar{L}_i K_i}{\rho_i \lambda_\infty} \sum_{j \geq 1} \beta_i^j u_i(k + j).$$

Let $c_i := \frac{\rho_i \lambda_\infty}{\bar{L}_i K_i}$ and define

$$S_k := \sum_{m \geq k} \beta_i^m u_i(m).$$

Note that

$$S_{k+1} = \sum_{m \geq k+1} \beta_i^m u_i(m) = \beta_i^k \sum_{m \geq 1} \beta_i^m u_i(k+m) \leq c_i \beta_i^k u_i(k), \quad (23)$$

and $S_k = \beta_i^k u_i(k) + S_{k+1}$, which gives

$$\beta_i^k u_i(k) = S_k - S_{k+1}. \quad (24)$$

Using (23) and (24), we obtain

$$S_{k+1} \leq c_i(S_k - S_{k+1}) \implies (1 + c_i)S_{k+1} \leq c_i S_k \implies S_k \leq S_2 \left(\frac{c_i}{1 + c_i} \right)^{k-2}.$$

Furthermore, since the eigenvector equations imply $\beta_i^k u_i(k) \leq S_k$, we obtain

$$u_i(k) \leq \frac{S_2}{\beta_i} \left(\frac{c_i}{(1 + c_i)\beta_i} \right)^{k-2}.$$

Therefore, by the root test for the radius of convergence,

$$\limsup_{k \rightarrow \infty} u_i(k)^{1/k} \leq \frac{c_i}{(1 + c_i)\beta_i} \implies R_i \geq \frac{(1 + c_i)\beta_i}{c_i} > \beta_i.$$

Since $R_i > \beta_i$ for every $i \in [N]$, we also have

$$\lambda_\infty u_j(k) \geq \frac{\bar{L}_j K_j}{\rho_j} \beta_j u_j(k).$$

This implies that the corresponding power series have the same radius of convergence. Hence, $R_i = R_j$ for all $i, j \in [N]$. Therefore, $R_i > \beta_{\max}$ for all $i \in [N]$, as desired. $\square$

By Lemma 11, the generating function $S(z)$ also has radius of convergence $R_S > \beta_{\max}$. From now on, we write R for the common radius of convergence of U_i and S , since they share the same radius of convergence. We now aim to obtain a more explicit representation of S using the fixed-point equation for $\mathcal{T}_{\lambda_\infty}$.

We recall that the fixed points of the limiting eigenvector equation satisfies the following for $k \geq 2$:

$$\lambda_\infty u_i(k) = \underbrace{\left(K_i + \frac{\bar{L}_i K_i}{\rho_i} \right) S(k-1)}_I + \underbrace{(\bar{K}_i - K_i) u_i(k-1)}_{II} + \underbrace{\frac{\bar{L}_i K_i}{\rho_i} \sum_{m \geq 1} \beta_i^m S(k+m-1)}_{III}. \quad (25)$$

The LHS of (25) can be used to obtain the following generating function

$$\lambda_\infty \sum_{k \geq 2} u_i(k) = \lambda_\infty (U_i(z) - u_i(1)z). \quad (26)$$

Similarly, using the term I , we obtain the generating function

$$\sum_{k \geq 2} \left(K_i + \frac{\bar{L}_i K_i}{\rho_i} \right) S(k-1) z^k = \left(K_i + \frac{\bar{L}_i K_i}{\rho_i} \right) z S(z), \quad (27)$$

which trivially has the same radius of convergence R as S and U_i . Using the term II , we obtain

$$\sum_{k \geq 2} (\bar{K}_i - K_i) u_i(k-1) z^k = (\bar{K}_i - K_i) z U_i(z), \quad (28)$$

which also has the radius of convergence R . It remains to construct a generating function for the term *III*.

First, define

$$y_i(k) = \sum_{m \geq 1} \beta_i^m s(k+m-1).$$

We use $y_i(k)$ to construct the generating function

$$Y_i(z) = \sum_{k \geq 2} y_i(k) z^k = \sum_{k \geq 2} \sum_{m \geq 1} \beta_i^m s(k+m-1) z^k. \quad (29)$$

Note that for each $i \in [N]$, the radius of convergence of Y_i about $z = 0$ is R . This can be deduced from (25), since terms *I* and *II* induce generating functions with radius of convergence R , as discussed above.

Thus, $Y_i(z)$ is analytic in a neighborhood of the origin, specifically on the open disk of radius $R > \beta_{\max}$ around $z = 0$. Our aim is to show that S admits a meromorphic extension to the whole complex plane, which will imply that the poles of S on the complex plane must have a finite order. To this end, we first provide a more explicit representation for Y_i on its domain of analyticity by arguing through algebraic manipulations and analytic extensions, which will then be used to recover an explicit representation for S in its radius of convergence.

Lemma 12. *For all $|z| < R$, we have*

$$Y_i(z) = \frac{\beta_i z}{z - \beta_i} (S(z) - S(\beta_i)).$$

Proof. Note that on the annulus $R > |z| > \beta_{\max}$, applying the Fubini-Tonelli theorem to (29) (which is applicable due to Lemma 11) we

$$Y_i(z) = \sum_{m \geq 1} \beta_i^m \sum_{k \geq 2} s(k+m-1) z^k \quad (30)$$

$$= \sum_{m \geq 1} \beta_i^m \sum_{j \geq m+1} s(j) z^{j+1-m} \quad (31)$$

$$= \sum_{m \geq 1} \frac{\beta_i}{z^m} z \left(S(z) - \sum_{n=1}^{m-1} s(n) z^n \right) \quad (32)$$

$$= \frac{\beta_i}{1 - \frac{\beta_i}{z}} S(z) - \sum_{m \geq 1} \beta_i^m z^{1-m} \sum_{n=1}^{m-1} s(n) z^n. \quad (33)$$

Define

$$\tilde{Y}_i(z) = \frac{\beta_i}{1 - \frac{\beta_i}{z}} S(z) - \sum_{m \geq 1} \beta_i^m z^{1-m} \sum_{n=1}^{m-1} s(n) z^n,$$

which is analytic on the annulus $R > |z| > \beta_i$ and on the disc $|z| < \beta_i$. As shown above, by the identity theorem we have $\tilde{Y}_i(z) = Y_i(z)$ on $R > |z| > \max_j \beta_j$. Since $Y_i(z)$ is analytic on $|z| < R$, it follows that $|z| = \beta$ is not a singularity region for $\tilde{Y}_i(z)$. Indeed, on the disc $|z| < R$, $\tilde{Y}_i(z)$ admits the analytic extension

$$\hat{Y}_i(z) = \left(\frac{\beta_i}{1 - \frac{\beta_i}{z}} S(z) - \sum_{m \geq 1} \beta_i^m z^{1-m} \sum_{n=1}^{m-1} s(n) z^n \right) 1_{\{R > |z| > \beta_i\}} + Y_i(z) 1_{\{|z| \leq \beta_i\}}.$$

Since $\hat{Y}_i(z)$ is analytic on $|z| < \beta_i$, it follows that $\hat{Y}_i(z)$ is an analytic extension of $\tilde{Y}_i(z)$ to the disc $|z| < R$. Indeed, by definition, we have $\hat{Y}_i(z) = \tilde{Y}_i(z)$ on the annulus $\max_j \beta_j < |z| < R$. Thus, it follows that $\hat{Y}_i(z) = \tilde{Y}_i(z)$ on $|z| < R$ by the identity theorem. To show this, note that,

using the fact that $S(z)$ is analytic on the annulus $0 < |z - \beta_i| < \varepsilon$, we obtain that $z = \beta_i$ is a removable singularity for $\tilde{Y}_i(z)$. Then, since $\tilde{Y}_i(z)$ is also analytic on $|z| < \max_j \beta_j$, it follows that $\tilde{Y}_i(z) = Y_i(z)$ on $|z| < R$ for all $i \in [N]$ by the identity theorem (after canceling the removable singularities). Thus, $\hat{Y}_i(z) = \tilde{Y}_i(z)$ on $|z| < R$.

It remains to control the term $\sum_{m \geq 1} \beta_i^m z^{1-m} \sum_{n=1}^{m-1} S(n) z^n$. Note that when $|z| > \beta_i$, we have

$$\begin{aligned} \sum_{m \geq 1} \beta_i^m z^{1-m} \sum_{n=1}^{m-1} s(n) z^n &= \sum_{m \geq 2} \sum_{n=1}^{m-1} \beta_i^m s(n) z^{1-m+n} = \sum_{n \geq 1} s(n) \sum_{m \geq n+1} \beta_i^m z^{n+1-m} \\ &= \sum_{n \geq 1} s(n) \sum_{t \geq 0} \beta_i^{t+n+1} z^{-t} = \sum_{n \geq 1} s(n) \beta_i^{n+1} \sum_{t \geq 0} \left(\frac{\beta_i}{z} \right)^t \\ &= \sum_{n \geq 1} s(n) \beta_i^{n+1} \frac{1}{1 - \beta_i/z} = \frac{\beta_i}{1 - \beta_i/z} S(\beta_i). \end{aligned}$$

Using the same analytic continuation argument above, we then obtain

$$Y_i(z) = \frac{\beta_i z (S(z) - S(\beta_i))}{z - \beta_i}$$

on the whole $|z| < R$, as desired. $\square$

Using generating functions to rewrite (25), we obtain

$$(\lambda_\infty - (\bar{K}_i - K_i)z) U_i(z) = \lambda_\infty u_i(1)z + \left(K_i + \frac{\bar{L}_i K_i}{\rho_i} \right) z S(z) + \frac{\bar{L}_i K_i}{\rho_i} Y_i(z). \quad (34)$$

Let

$$H_i(z) := \frac{\lambda_\infty u_i(1)z + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i z}{z - \beta_i} (S(z) - S(\beta_i))}{\lambda_\infty - (\bar{K}_i - K_i)z}, \quad B_i(z) := \frac{(K_i + \frac{\bar{L}_i K_i}{\rho_i})z}{\lambda_\infty - (\bar{K}_i - K_i)z}. \quad (35)$$

Note that $H_i(z)$ is analytic over $|z| < R$ by Lemma 12. Next, we prove that S admits a meromorphic extension, which in turn implies that the only singularities of S are poles. Furthermore, we will identify an upper bound for the radius of convergence R . We will need these explicit bounds on R to relate λ_∞ with some $\rho(B(r))$. Before proceeding, for technical purposes, we need the inequality $\frac{\lambda_\infty}{\bar{K}_i - K_i} > \beta_i$.

Lemma 13. *For all $i \in [N]$, we have*

$$\lambda_\infty \geq \bar{K}_i \beta_i.$$

In particular, we have

$$\lambda_\infty \geq \frac{1}{2} \frac{\bar{L}_i K_i \beta_i}{\rho_i}.$$

Proof. Each diagonal block matrix in the definition of $\mathcal{S}_T$ amounts to a contraction condition for the single population case. Due to Perron-Frobenius monotonicity, for all $i \in \mathbb{N}$, it holds that $\rho(\mathcal{S}_T) \geq \rho(\mathcal{M}_i)$. By [6, Theorem 2], we have $\rho(\mathcal{M}_i) \geq \bar{K}_i \beta_i$, so the desired result holds. $\square$

Lemma 14. *S admits a meromorphic extension and on $|z| < R$ it satisfies*

$$S(z) = \frac{\sum_{i \in [N]} \frac{\lambda_\infty u_i(1)z - \frac{\bar{L}_i K_i \beta_i z}{\rho_i(z - \beta_i)} S(\beta_i)}{\lambda_\infty - (\bar{K}_i - K_i)z}}{1 - \sum_{i \in [N]} \frac{(K_i + \frac{\bar{L}_i K_i}{\rho_i})z + \frac{\bar{L}_i K_i \beta_i z}{\rho_i(z - \beta_i)}}{\lambda_\infty - (\bar{K}_i - K_i)z}}. \quad (36)$$

In particular, singularities of S are either removable or a pole of finite order. Furthermore, $R \leq \max_i \frac{\lambda_\infty}{\bar{K}_i - K_i}$.

Proof. (34) implies that

$$U_i(z) = H_i(z) + B_i(z)S(z). \quad (37)$$

Using the definition of S and (37), we can express S as

$$S(z) = \sum_{i \in [N]} U_i(z) = \sum_{i \in [N]} \left(\frac{\lambda_\infty u_i(1)z - \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i z}{z - \beta_i} S(\beta_i)}{\lambda_\infty - (\bar{K}_i - K_i)z} + S(z) \frac{(K_i + \frac{\bar{L}_i K_i}{\rho_i})z + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i z}{z - \beta_i}}{\lambda_\infty - (\bar{K}_i - K_i)z} \right). \quad (38)$$

It follows that (36) holds on $|z| < R$. Thus, S admits a meromorphic extension by the identity theorem.

It remains to show that $R \leq \max_i \frac{\lambda_\infty}{\bar{K}_i - K_i}$. Note that $\frac{\bar{L}_i K_i}{\rho_i} S(\beta_i) = \lambda_\infty u_i(1)$ as a consequence of the boundary conditions of $\mathcal{T}_{\lambda_\infty}$, see (13). Thus,

$$\lambda_\infty u_i(1)z - \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i z}{z - \beta_i} S(\beta_i) = \frac{\lambda_\infty u_i(1)z^2}{z - \beta_i}.$$

Since $S(x) > 0$ for any $x > 0$, note that for any $x \in (\beta_{\max}, \max_i \frac{\lambda_\infty}{\bar{K}_i - K_i})$, (38) implies that

$$S(x) \geq \sum_{i \in [N]} \frac{\lambda_\infty u_i(1)x^2}{(x - \beta_i)(\lambda_\infty - (\bar{K}_i - K_i)x)}.$$

By Lemma 13, the limit

$$\lim_{x \rightarrow (\max_i \frac{\lambda_\infty}{\bar{K}_i - K_i})^-} \sum_{i \in [N]} \frac{\lambda_\infty u_i(1)x^2}{(x - \beta_i)(\lambda_\infty - (\bar{K}_i - K_i)x)} = \infty \quad (39)$$

holds; therefore, we must have $R \leq \max_i \frac{\lambda_\infty}{\bar{K}_i - K_i}$, where the limit in (39) is taken over the real line. $\square$

Next, using the characterization of the poles of S and the existence of the eigenvector $(u_i(k))_{i,k}$, we prove that the “smallest” pole of S must satisfy a relation with λ_∞ similar to the relation presented between $\rho(B(r))$ and r in Lemma 10.

Proposition 3. *There exists $z_0 \in (\beta_{\max}, \max_i \frac{\lambda_\infty}{\bar{K}_i - K_i})$ such that*

$$1 = \sum_{i \in [N]} \frac{(K_i + \frac{\bar{L}_i K_i}{\rho_i})z_0 + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i z_0}{z_0 - \beta_i}}{\lambda_\infty - (\bar{K}_i - K_i)z_0}.$$

Furthermore, z_0 is a simple pole of S .

Proof. By Lemma 11, the singularities $z = \beta_i$ of S , $i \in [N]$, are removable. Thus, all non-removable singularities of S come from z such that $1 = \sum_{i \in [N]} \frac{(K_i + \frac{\bar{L}_i K_i}{\rho_i})z + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i z}{z - \beta_i}}{\lambda_\infty - (\bar{K}_i - K_i)z}$. We recall that for a power series of the form $f(z) = \sum_n a_n z^n$, Vivanti-Pringsheim theorem says that if the radius of convergence of f is $R < \infty$, and $a_n \geq 0$ for all n , then $z = R$ is a singularity of f . Since $(u_i(k))_{i,k}$ are all positive, using the definition (22) and Vivanti-Pringsheim theorem, we see that S must have a singularity at the boundary $z = R$, which corresponds to a value that satisfies $1 = \sum_{i \in [N]} \frac{(K_i + \frac{\bar{L}_i K_i}{\rho_i})z + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i z}{z - \beta_i}}{\lambda_\infty - (\bar{K}_i - K_i)z}$ when $R < \infty$.

First, we will show that in a disc with radius $R < \infty$, there is a non-removable singularity of S . Indeed, the equation (39) in Lemma 14 shows that $z = \max_i \frac{\lambda_\infty}{\bar{K}_i - K_i}$ is a non-removable singularity of S , by definition. Thus, we must have $R < \infty$, meaning that there exists $z_0 \in (0, \infty)$ such that

$$1 = \sum_{i \in [N]} \frac{(K_i + \frac{\bar{L}_i K_i}{\rho_i})z_0 + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i z_0}{z_0 - \beta_i}}{\lambda_\infty - (\bar{K}_i - K_i)z_0} \quad (40)$$

that is not canceled by the numerator of $S(z)$. Note that (40) must hold as by Lemma 11, we also must have $z_0 > \beta_{\max}$, meaning that the numerator of S does not have any singularity. The upper bound on z_0 follows from Lemma 14. The upper bound of z_0 is not sharp since (40) is not feasible at the boundary $z = \max_i \frac{\lambda_\infty}{\bar{K}_i - K_i}$.

For the last claim, since the contributions to the singularities that come from the denominator of S is a rational function made of order one polynomials, the pole must be simple. $\square$

Proposition 3 shows that if $\mathcal{T}_{\lambda_\infty}$ admits a positive fixed-point u , then

$$1 = \sum_{i \in [N]} \frac{K_i z_0 + \frac{\bar{L}_i K_i}{\rho_i} \frac{z_0}{z_0 - \beta_i}}{\lambda_\infty - (\bar{K}_i - K_i) z_0}$$

must hold for λ_∞ for some z_0 . Next, we will prove the converse, i.e. if λ_∞ satisfies

$$1 = \sum_{i \in [N]} \frac{K_i z_0 + \frac{\bar{L}_i K_i}{\rho_i} \frac{z_0}{z_0 - \beta_i}}{\lambda_\infty - (\bar{K}_i - K_i) z_0}$$

for some $z_0 \in (\beta_{\max}, \infty)$, then there exists a positive fixed point for $\mathcal{T}_{\lambda_\infty}$. For this purpose, we will rely on standard fixed-point arguments and define a compact domain for $\mathcal{T}_{\lambda_\infty}$.

This procedure will rely on an intermediate step to relate λ_∞ with $\rho(B(r))$ for some r . To be more precise, our aim will be to prove that

$$\lim_{k \rightarrow \infty} \frac{u_i(k+1)}{u_i(k)} = \frac{1}{z_0} \quad \text{for each } i \in [N], \quad (41)$$

where z_0 is the same for all $i \in [N]$. The proof will solely rely on singularity analysis, so first, we recall some basic results from complex analysis that will be essential in proving (41).

Lemma 15 (Cauchy's coefficient formula). *Let f be analytic on $\{z : |z| \leq r\}$. By $[z^k]f(z) = a^k$ denote the coefficient of the term z^k on the power series representation of $f = \sum_n a_n z^n$. Then for each $k \geq 0$,*

$$[z^k]f(z) = \frac{1}{2\pi i} \int_{|z|=r} \frac{f(z)}{z^{k+1}} dz.$$

Proof. See [2, Chapter 4, Section 2, Lemma 3]. $\square$

A well known consequence of Lemma 15 is the following boundary estimate, which is also due to Cauchy:

Lemma 16. *If f is analytic on $\{|z| \leq r\}$ and $M_r := \max_{|z|=r} |f(z)|$, then*

$$|[z^k]f(z)| \leq \frac{M_r}{r^k} \quad (k \geq 0).$$

Proof. Apply Lemma 15 and bound:

$$|[z^k]f(z)| \leq \frac{1}{2\pi} \int_{|z|=r} \frac{|f(z)|}{|z|^{k+1}} |dz| \leq \frac{1}{2\pi} \int_{|z|=r} \frac{M_r}{r^{k+1}} |dz| = \frac{M_r}{r^k}.$$

$\square$

In Lemma 14, we showed that S only admits poles as non-removable singularities. To obtain (41), we will first obtain this result for the bulk eigenvector term by using basic singularity analysis by using the tools mentioned above and the residue theorem.

Lemma 17. *There exist constants $c \neq 0$ and $C_{R+\varepsilon} > 0$ such that*

$$s(k) = cz_0^{-k} + O((R+\varepsilon)^{-k}), \quad |s(k) - cz_0^{-k}| \leq C_{R+\varepsilon}(R+\varepsilon)^{-k}.$$

Consequently,

$$\frac{s(k+1)}{s(k)} \rightarrow \frac{1}{z_0}.$$

Proof. By Proposition 3, S has a simple pole at $z_0 = R$ isolated from others. Thus, by Lemma 14, there exists $c \neq 0$ and an analytic function G on $|z| < R + \varepsilon$ such that

$$S(z) = \frac{c}{z_0 - z} + G(z). \quad (42)$$

Fix the complex circle $|z| = (R + \varepsilon)$ (with $R + \varepsilon > z_0$). By Proposition 3, we can choose $\varepsilon > 0$ so that the only singularity of S inside the disc $|z| \leq R + \varepsilon$ is z_0 . Then, using Lemma 15, we obtain

$$s(k) = [z^k]S(z) = \frac{1}{2\pi i} \int_{|z|=R+\varepsilon} \frac{S(z)}{z^{k+1}} dz.$$

Insert (42):

$$s(k) = \frac{1}{2\pi i} \int_{|z|=R+\varepsilon} \frac{c}{(z_0 - z)z^{k+1}} dz + \frac{1}{2\pi i} \int_{|z|=R+\varepsilon} \frac{G(z)}{z^{k+1}} dz.$$

The first integral is computed by the residue theorem [2, Theorem 17]: the only pole inside $|z| = R + \varepsilon$ is at $z = z_0$, and its residue equals cz_0^{-k} . Hence

$$\frac{1}{2\pi i} \int_{|z|=R+\varepsilon} \frac{c}{(z_0 - z)z^{k+1}} dz = cz_0^{-k}.$$

For the second integral, apply Lemma 16 to G (analytic on $|z| \leq R + \varepsilon$):

$$\left| \frac{1}{2\pi i} \int_{|z|=R+\varepsilon} \frac{G(z)}{z^{k+1}} dz \right| = |[z^k]G(z)| \leq \frac{M_{R+\varepsilon}(G)}{(R+\varepsilon)^k}.$$

Thus $s(k) = cz_0^{-k} + O((R+\varepsilon)^{-k})$.

For the ratio, write

$$\frac{s(k+1)}{s(k)} = \frac{cz_0^{-(k+1)} + O((R+\varepsilon)^{-(k+1)})}{cz_0^{-k} + O((R+\varepsilon)^{-k})} = \frac{1}{z_0} \cdot \frac{1 + O((z_0/(R+\varepsilon))^{k+1})}{1 + O((z_0/(R+\varepsilon))^k)} \rightarrow \frac{1}{z_0},$$

as $k \rightarrow \infty$ since $z_0/(R+\varepsilon) < 1$ and the leading coefficients of the terms $O((z_0/(R+\varepsilon))^{k+1})$ and $O((z_0/(R+\varepsilon))^k)$ are the same. $\square$

Next, using equation (34), we will use a similar singularity analysis on $U_i(z)$ through $S(z)$ to obtain (41) since Lemma 14 implies that the singularities of $U_i(z)$ must come from $S(z)$. We will also identify an asymptotic cross-interaction between the terms $u_i(k)$ and $u_j(k)$ between different populations that will be useful while identifying z_0 such that $\lambda_\infty = \rho(B(1/z_0))$.

Lemma 18. *There exist $c_i \neq 0$ and $C_{i,R+\varepsilon} > 0$ such that*

$$u_i(k) = c_i z_0^{-k} + O((R+\varepsilon)^{-k}), \quad |u_i(k) - c_i z_0^{-k}| \leq C_{i,R+\varepsilon}(R+\varepsilon)^{-k},$$

and consequently $\lim_{k \rightarrow \infty} \frac{u_i(k+1)}{u_i(k)} = \frac{1}{z_0}$. Furthermore, there exist positive constants $c_i, c_j > 0$ such that

$$\lim_{k \rightarrow \infty} \frac{u_i(k)}{u_j(k)} = \frac{c_i}{c_j}$$

for all $i, j \in [N]$.

Proof. Using (25) and (35), we have

$$U_i(z) = \frac{\lambda_\infty u_i(1)z^2}{(\lambda_\infty - (\bar{K}_i - K_i)z)(z - \beta_i)} + S(z) \left(\frac{(K_i + \frac{\bar{L}_i K_i}{\rho_i})z + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i z}{z - \beta_i}}{\lambda_\infty - (\bar{K}_i - K_i)z} \right).$$

It follows that the singularity $z_0 \in (\beta_{\max}, \frac{\lambda_\infty}{\max_i(\bar{K}_i - K_i)})$ obtained for S in Proposition 3 is also the singularity of $U_i(z)$. Using (42), and Lemma 14, we obtain

$$U_i(z) = \frac{\lambda_\infty u_i(1)z^2}{(\lambda_\infty - (\bar{K}_i - K_i)z)(z - \beta_i)} + \frac{c \left((K_i + \frac{\bar{L}_i K_i}{\rho_i})z_0 + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i z_0}{z_0 - \beta_i} \right)}{(z - z_0)(\lambda_\infty - (\bar{K}_i - K_i)z_0)} + G_i(z),$$

for some $G_i(z)$ that is analytic in the region $|z| < \frac{\lambda_\infty}{\max_i(\bar{K}_i - K_i)}$. We let

$$c_i = \frac{c \left((K_i + \frac{\bar{L}_i K_i}{\rho_i})z_0 + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i z_0}{z_0 - \beta_i} \right)}{\lambda_\infty - (\bar{K}_i - K_i)z_0} > 0.$$

Then, there exists some $\tilde{G}_i$ that is analytic in the region $|z| < \frac{\lambda_\infty}{\max_i(\bar{K}_i - K_i)}$ such that

$$U_i(z) = \frac{c_i}{z - z_0} + \tilde{G}_i(z).$$

Now repeat the proof of Lemma 17 with U_i in place of S :

$$u_i(k) = [z^k]U_i(z) = c_i z_0^{-k} + [z^k]\tilde{G}_i(z),$$

and by Lemma 16,

$$|[z^k]\tilde{G}_i(z)| \leq \frac{M_{R+\varepsilon}(\tilde{G}_i)}{(R+\varepsilon)^k}.$$

Hence $u_i(k) = c_i z_0^{-k} + O((R+\varepsilon)^{-k})$, and the ratio limit follows:

$$\frac{u_i(k+1)}{u_i(k)} = \frac{c_i z_0^{-(k+1)} + O((R+\varepsilon)^{-(k+1)})}{c_i z_0^{-k} + O((R+\varepsilon)^{-k})} \rightarrow \frac{1}{z_0}.$$

Similarly, for $i \neq j$, the cross ratio term

$$\lim_{k \rightarrow \infty} \frac{u_i(k)}{u_j(k)} = \lim_{k \rightarrow \infty} \frac{c_i z_0^{-k} + O((R+\varepsilon)^{-k})}{c_j z_0^{-k} + O((R+\varepsilon)^{-k})} = \frac{c_i}{c_j}$$

also follows from the same argument. □

By Lemma 18, for every i ,

$$\lim_{k \rightarrow \infty} \frac{u_i(k+1)}{u_i(k)} = \frac{1}{z_0}.$$

Define $r := 1/z_0$. Then $u_i(k+1)/u_i(k) \rightarrow r$ for all i , i.e. the ratio tends to a constant independent of i . The same also applies to the asymptotic cross-interaction between different population groups. Next, we will show that this implies that $\lambda_\infty = \rho(B(r))$. This allow us to relate Proposition 3 to Lemma 8.

Lemma 19. *We have $\lambda_\infty = \rho(B(1/z_0))$.*

Proof. For all $k \geq 2$, using the positive fixed point of limiting eigenvector equation $\mathcal{T}_{\lambda_\infty}$, which exists as shown in Proposition 2, we obtain

$$\lambda_\infty = (\bar{K}_i - K_i) \frac{u_i(k)}{u_i(k+1)} + \left(K_i + \frac{\bar{L}_i K_i}{\rho_i} \right) \sum_{j \in [N]} \frac{u_j(k)}{u_i(k)} + \frac{\bar{L}_i K_i}{\rho_i} \sum_{j \in [N]} \sum_{m \geq 1} \beta_i^m \frac{u_j(k+m)}{u_i(k+1)}.$$

Using Lemma 18, and the Fatou lemma, we then obtain that

$$\lambda_\infty [c_1 \ c_2 \ c_3 \ \cdots \ c_N]^\top \geq ([c_1 \ c_2 \ c_3 \ \cdots \ c_N] B(1/z_0)^\top)^\top,$$

where the constants c_i are the ones specified in Lemma 18. Since $z_0 > 0$, $c_i > 0$ and $\lambda_\infty > 0$, as a consequence of the Perron-Frobenius theorem, we obtain that $\rho(B(1/z_0)) \leq \lambda_\infty \leq \inf_{z \geq \beta_{\max}} \rho(B(1/z))$, and thus it holds that $\rho(B(1/z_0)) = \lambda_\infty$, as desired. $\square$

Proof of Proposition 1. The result follows from Lemmas 7 and 19. $\square$

Proof of Theorem 5. Throughout this subsection, we did not use any property specific to the constant λ_∞ . Therefore, by Lemma 19, the same conclusion remains valid whenever $\mathcal{T}_\lambda$ admits a positive fixed point for some $\lambda > 0$. $\square$

Lemma 19 technically presents a complete characterization when $\lambda_\infty < 1$, so we have completed the proof of one side of Theorem 4 as expressed in Proposition 1. However, Proposition 1 still requires calculating $\inf_r \rho(B(r))$, which can be inconvenient, especially when N is large as the matrix $B(r)$ is not sparse. Indeed, standard methods in the literature to calculate $\rho(B(r))$ requires $O(N^3)$ operations.

To complete the proof of the other half of Theorem 4, we will need the following auxiliary lemma, which will allow us to directly relate λ values for which $\mathcal{T}_\lambda$ admits a fixed point and the values $\rho(B(r))$.

Lemma 20. Fix $t > C := \max_{1 \leq i \leq n} c_i$ and define

$$N_i(t) := a_i t + \frac{b_i t}{t - c_i} \quad \text{and} \quad p_i(t) := d_i t.$$

Let

$$p_m := \max_{1 \leq i \leq n} p_i(t).$$

For $e \in \mathbb{R} \setminus \{p_1(t), \dots, p_n(t)\}$, define

$$G(t, e) := \sum_{i=1}^n \frac{N_i(t)}{e - p_i(t)}.$$

Then, there exists a unique $e^* \in (p_m(t), \infty)$ such that

$$G(t, e^*) = 1.$$

Proof. Since $t > C \geq c_i$ for all i and $a_i, b_i, c_i > 0$, we have $t - c_i > 0$ and hence

$$N_i(t) = a_i t + \frac{b_i t}{t - c_i} > 0 \quad \text{for all } i.$$

If $e > d_m t$, then for every i ,

$$e - d_i t \geq e - d_m t > 0,$$

so each summand $\frac{N_i(t)}{e - d_i t}$ is finite. Therefore, $e \mapsto F(t, e)$ is well defined and continuous on $(d_m t, \infty)$.

For $e > d_m t$, we may differentiate term-by-term to obtain

$$\frac{\partial}{\partial e} \left(\frac{N_i(t)}{e - d_i t} \right) = -\frac{N_i(t)}{(e - d_i t)^2} \implies \frac{\partial G}{\partial e}(t, e) = \sum_{i=1}^n -\frac{N_i(t)}{(e - d_i t)^2}.$$

Since $N_i(t) > 0$ and $(e - d_i t)^2 > 0$ for all $e > d_m t$, we have

$$\frac{\partial G}{\partial e}(t, e) < 0 \quad \text{for all } e > d_m t.$$

Hence $G(t, \cdot)$ is strictly decreasing on (p_m, ∞) .

Let $I := \{i : d_i t = d_m t\}$, which is nonempty by definition of $d_m t$. As $e \rightarrow (d_m t)^+$, for each $i \in I$ we have $e - d_i t = e - d_m t \rightarrow 0^+$, and thus

$$\frac{N_i(t)}{e - d_i t} \rightarrow +\infty.$$

Since all terms are nonnegative for $e > d_m t$, it follows that

$$\lim_{e \rightarrow d_m t^+} G(t, e) = +\infty.$$

On the other hand, as $e \rightarrow \infty$ we have for each fixed i ,

$$\frac{N_i(t)}{e - d_i t} \rightarrow 0,$$

and therefore

$$\lim_{e \rightarrow \infty} G(t, e) = 0.$$

We have shown that the function $G(t, \cdot)$ is continuous on (p_m, ∞) and satisfies

$$\lim_{e \rightarrow (d_m t)^+} G(t, e) = +\infty \quad \text{and} \quad \lim_{e \rightarrow \infty} G(t, e) = 0.$$

Hence, by the intermediate value theorem, there exists $e^* \in (d_m t, \infty)$ such that $G(t, e^*) = 1$.

As we have shown that $G(t, \cdot)$ is strictly decreasing on $(d_m t, \infty)$, it can take the value 1 at most once on this interval. Therefore the solution e^* is unique in this case. $\square$

Next, we will prove a result that will amount to the proof of Theorem 4. This will amount to a quantitative condition that is equivalent to $\inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r)) < 1$, for which Lemma 20 will be crucial.

Lemma 21. *We have $\inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r)) < 1$ if and only if*

$$\inf_{\min_j (\bar{K}_j - K_j) < r < \beta_{\max}^{-1}} \sum_{i \in [N]} \frac{\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i} \frac{1}{1 - \beta_i r}}{1 - \frac{\bar{K}_i - K_i}{r}} \leq 1. \quad (43)$$

Nonemptiness of the interval $(\min_j (\bar{K}_j - K_j), \beta_{\max}^{-1})$ is automatic when $\inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r)) < 1$, i.e. it is part of the contraction criterion.

Proof. Suppose that it holds that $\inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r)) < 1$. First, we suppose that there exists a convergent sequence $(r_n)_n \subset (0, \beta_{\max}^{-1})$ such that $(\rho(B(r_n)))_n < 1$ for all $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} \rho(B(r_n)) = 1$. Then, we have that

$$1 \geq \inf_{\frac{\min_j (\bar{K}_j - K_j)}{\rho(B(r_n))} < r < \beta_{\max}^{-1}} \sum_{i \in [N]} \frac{\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i} \frac{1}{1 - \beta_i r}}{\rho(B(r_n)) - \frac{\bar{K}_i - K_i}{r}} > \inf_{\frac{\min_j (\bar{K}_j - K_j)}{\rho(B(r_n))} < r < \beta_{\max}^{-1}} \sum_{i \in [N]} \frac{\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i} \frac{1}{1 - \beta_i r}}{1 - \frac{\bar{K}_i - K_i}{r}}.$$

Taking the limit as $n \rightarrow \infty$, the inequalities above imply that

$$\inf_{\min_j (\bar{K}_j - K_j) \leq r \leq \beta_{\max}^{-1}} \sum_{i \in [N]} \frac{\frac{K_i}{r} + \frac{\bar{L}_i K_i}{\rho_i} \frac{1}{1 - \beta_i r}}{1 - \frac{\bar{K}_i - K_i}{r}} < 1,$$

as desired.

If no such sequence exists, i.e., $\sup_{0 < r < \beta_{\max}^{-1}} \rho(B(r)) < 1$, then for $r = 1$, it must hold that

$$1 = \sum_{i \in [N]} \frac{K_i + \frac{\bar{L}_i K_i}{\rho_i} + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i}{1 - \beta_i}}{\rho(B(1)) - (\bar{K}_i - K_i)} > \sum_{i \in [N]} \frac{K_i + \frac{\bar{L}_i K_i}{\rho_i} + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i}{1 - \beta_i}}{1 - (\bar{K}_i - K_i)}.$$

Note that for any positive Perron right eigenvector c of $B(1)$, we have

$$B(1)c = \rho(B(1))c < c,$$

meaning that $1 > \bar{K}_i - K_i$ for all $i \in [N]$ as $\rho(B(1)) \geq \bar{K}_i - K_i$ for all $i \in [N]$. Thus, we must have

$$\inf_{\min_j (\bar{K}_j - K_j) < r < \beta_{\max}^{-1}} \sum_{i \in [N]} \frac{\frac{K_i + \frac{\bar{L}_i K_i}{\rho_i}}{r} + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i}{1 - \beta_i r}}{1 - \frac{\bar{K}_i - K_i}{r}} < 1,$$

as desired.

To prove the converse, we suppose that

$$\inf_{\min_j (\bar{K}_j - K_j) < r < \beta_{\max}^{-1}} \sum_{i \in [N]} \frac{\frac{K_i + \frac{\bar{L}_i K_i}{\rho_i}}{r} + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i}{1 - \beta_i r}}{1 - \frac{\bar{K}_i - K_i}{r}} < 1. \quad (44)$$

Since

$$\lim_{r \rightarrow (\beta_{\max}^{-1})^-} \sum_{i \in [N]} \frac{\frac{K_i + \frac{\bar{L}_i K_i}{\rho_i}}{r} + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i}{1 - \beta_i r}}{1 - \frac{\bar{K}_i - K_i}{r}} = \lim_{r \rightarrow \min_j (\bar{K}_j - K_j)^+} \sum_{i \in [N]} \frac{\frac{K_i + \frac{\bar{L}_i K_i}{\rho_i}}{r} + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i}{1 - \beta_i r}}{1 - \frac{\bar{K}_i - K_i}{r}} = \infty,$$

the assumption (44) implies that there exists some r^{**} such that $\beta_{\max}^{-1} > r^{**} > \min_j (\bar{K}_j - K_j)$ and

$$\sum_{i \in [N]} \frac{\frac{K_i + \frac{\bar{L}_i K_i}{\rho_i}}{r^{**}} + \frac{\bar{L}_i K_i}{\rho_i} \frac{\beta_i}{1 - \beta_i r^{**}}}{1 - \frac{\bar{K}_i - K_i}{r^{**}}} = 1.$$

By Lemma 20, $\rho(B(r^{**})) = 1$; thus, it must hold that $1 > \lambda_\infty$ by Lemma 7, where the strict inequality is due to the fact that we can repeat the same argument for $1 - \varepsilon$ for sufficiently small $\varepsilon > 0$. $\square$

Proof of Theorem 4. The result follows directly from Lemma 21 as we have $\lambda_\infty = \rho(B(1/z_0))$ by Lemma 19, which implies

$$\lambda_\infty = \inf_{0 < r < \beta_{\max}^{-1}} \rho(B(r))$$

by Lemma 7, as desired. $\square$

In Section 5, we will use the value t_* that realizes the minimization problem

$$\inf_{0 < r < (\max_i \beta_i)^{-1}} V(r)$$

to construct an exponential weight vector and use a majorization argument (via Collatz-Wielandt) to obtain a uniqueness result for discounted infinite-horizon non-stationary MFE via approximation through (discounted) finite-horizon MFE. For this uniqueness result, we will need that $\lambda_\infty < 1$ and $t_* > 1$. Numerically, it is easy to find t_* that minimizes V over the desired interval; hence, for a given system one can check both of these conditions numerically.

Remark 2. In [6] (also see [7]), a similar contraction condition is obtained for the single population case. The proof present in [6, Section 3] fails in the multi-population case. Although the techniques presented here can be applied to obtain an equivalent upper bound for the contraction rate obtained in [6, Theorem 2], our techniques cannot describe a tight lower bound (that holds under any configuration) for $\rho(\mathcal{S}_T)$ when T is sufficiently large as done in [6, Corollary 2]. Furthermore, the proof of [6, Theorem 2] indicates that in the single population group case the left eigenvectors of $\mathcal{S}_T$ have a constant ratio limit under normalization, which was proved only for the asymptotics of the ratios.

5 Finite-Horizon MFE to Infinite-Horizon Non-Stationary MFE: Convergence Rates and Uniqueness

Let

$$(X, \mathcal{P}(A), (C_i + \rho_i \Omega_i)_{i \in [N]}, (P_i)_{i \in [N]}, N)$$

be a regularized multi-population MFG in which the objective function of each population $i \in [N]$ is discounted by the factor β_i . In this section, we will establish a uniqueness result for infinite-horizon non-stationary MFE and a finite-time error bound on the convergence of finite-horizon MFE to infinite-horizon non-stationary MFE. For this purpose, we will use the variational contraction function V to construct weights and use a majorization argument under these weights on the matrices $(\mathcal{S}_T)_T$.

For any $(\pi^\infty, \tau^\infty) \in \text{MFE}_{\tau_0}$, by $(\pi^\infty|_T, \tau^\infty|_T)$, we will denote the time truncation of this MFE to the first T -coordinates of every population group. The main result of this section is the following finite-time error bound, from which the uniqueness of the non-stationary MFE follows directly:

Theorem 6. *Let $(\pi^T, \tau^T) \in \text{MFE}_{T, \tau_0}$. Suppose that there exists $t_o > 1$ such that $V(t_o) < 1$. Then, for all $t \ll T$, we have*

$$\sum_{i \in [N]} \|\tau_{t,i}^\infty|_T - \tau_{t,i}^T\|_1 \leq O\left(\frac{1}{t_o^{T-t}}\right).$$

In particular, $|\text{MFE}_{\tau_0}| = 1$, i.e., there exists a unique non-stationary MFE in the N -population case under any given family of initial state-measure τ_0 .

Remark 3. *If $V(1) < 1$, then the working conditions of Theorem 6 are satisfied automatically due to the continuity of V . When $V(1) > 1$ and $V(t_*) > 1$, it is easy to see that the assumptions of Theorem 6 are not feasible. Thus, the interesting cases are when $V(1) > 1$ and $V(t_*) < 1$.*

The proof of Theorem 6 relies on using the function V to create a majorizing Lyapunov function as done in the proof of Lemma 7. The construction of this Lyapunov function relies on the minimizers of the function V used in Theorem 3 to characterize λ_∞ . The main difference in this case will be that we will majorize the transpose of $\mathcal{S}_T$ to obtain these error bounds.

The condition $\lambda_\infty < 1$ is required to control the differences $\tau_{t,i}^\infty|_T - \tau_{t,i}^T$ through any horizon length T , while we need $t_* > 1$ to control the contribution of the remainder terms that appear due to the fact that τ^∞ induces an infinite-horizon horizon MFE.

Let $\bar{t}_*(T) = (t_*^{T-2}, t_*^{T-3}, \dots, t_*, 1)$. By $c = (c_1, c_2, \dots, c_N)$ denote a positive left Perron eigenvector of $B(t_*)$ and set $C = \sum_{i \in [N]} c_i$. Using these, we define

$$\hat{t}_*(T) = \underbrace{(c_1 \bar{t}_*(T), c_2 \bar{t}_*(T), \dots, c_N \bar{t}_*(T))}_{N \text{ times}}.$$

For all $T \in \mathbb{N}$, we construct our Lyapunov function as $V(\tau, T) = \langle \hat{t}_*, \tau \rangle$. The motivation behind the cyclic behavior of $\hat{t}_*$ can be readily seen from the proof of Lemma 7, where we reduced the problem of finding a bound for $\rho(\mathcal{S}_T)$ to that of $\rho(B(r))$ using the special structure of eigenspace that corresponds to $\rho(B(r))$. To obtain Theorem 6, we start with a version of Lemma 7 but for the transpose of $\mathcal{S}_T^*$.

Lemma 22. *For all $n \in \mathbb{N}$, and $m \in [T-2] \cup \{0\}$, it holds that*

$$\max_i \frac{(\mathcal{S}_n^* \hat{t}_*)_{m(T-1)+i}}{c_i \bar{t}_{*,i}} \leq \rho(B(t_*)).$$

Proof. For $1 \leq i \leq n-1$ we have $\bar{t}_{*,i+1} = \bar{t}_{*,i}/t_*$, and, for $1 \leq k \leq i$, $\bar{t}_{*,i-k+1} = t_*^{k-1} \bar{t}_{*,i}$. A direct inspection as done in Lemma 7 gives

$$\frac{(\mathcal{S}_n^* \hat{t}_*(n))_{m(T-1)+i}}{c_m \bar{t}_{*,i}(n)} \leq \rho(B(t_*)).$$

The same also holds for $i = n$. Thus, for all i , the desired inequality holds after combining the cases above. $\square$

Lemma 23. *With the same notation of Lemma 22, for any positive vector $y \in \mathbb{R}^{N(T-1)}$, we have that*

$$\langle \hat{t}_*(T), \mathcal{S}_T y \rangle \leq \rho(B(t_*)) \langle \hat{t}_*(T), y \rangle.$$

Proof. By Lemma 22, we have

$$\mathcal{S}_T^* \hat{t}_*(T) \leq \begin{bmatrix} \rho(B(t_*)) c_1 \bar{t}_*(T) \\ \rho(B(t_*)) c_2 \bar{t}_*(T) \\ \vdots \\ \rho(B(t_*)) c_N \bar{t}_*(T) \end{bmatrix} = \rho(B(t_*)) \hat{t}_*(T), \quad (45)$$

where the last equality is due to Theorem 3. Since taking inner product with a positive vector y will preserve (45), the desired result follows. $\square$

In the proof of Theorem 6, we will denote $\hat{t}_*(T)$ by $\hat{t}_*$ for simplicity.

Proof of Theorem 6. If $t_o \neq t_*$, then we can perturb $\mathcal{S}_T$ with a positive multiple of the identity matrix so that its spectral radius is $V(t_o)$. Thus, without loss of generality, we will suppose that $t_* > 1$ for the remainder of this proof.

Let $\tau^T = (\tau_t^T)_{t=0}^{T-1}$ be a finite-horizon MFE obtained under the horizon length T starting from τ_0 . Then, under the Euclidean inner product $\langle \cdot, \cdot \rangle$, arguing as in Lemma 5, and using Lemma 23, we obtain that

$$\begin{aligned} & \langle \hat{t}_*, (\|\tau_{t,j}^\infty|_T - \tau_{t,j}^T\|_1)_{t=1,j=1}^{T-1,N} \rangle \\ & \leq \langle \hat{t}_*, \mathcal{S}_T (\|\tau_{t,j}^\infty|_T - \tau_{t,j}^T\|_1)_{t=1,j=1}^{T-1,N} \rangle + \langle \hat{t}_*, \bar{\mathcal{S}} \left(\underbrace{\|Q_T^{\tau^\infty,i}\|_\infty, \|Q_T^{\tau^\infty,i}\|_\infty, \dots, \|Q_T^{\tau^\infty,i}\|_\infty}_{T-1 \text{ times}} \right)_{i \in [N]} \rangle \\ & \leq \rho(B(t_*)) \langle \hat{t}_*, (\|\tau_{t,j}^\infty|_T - \tau_{t,j}^T\|_1)_{t=1,j=1}^{T-1,N} \rangle \\ & \quad + \langle \hat{t}_*, \bar{\mathcal{S}} \left(\|Q_T^{\tau^\infty,i}\|_\infty, \|Q_T^{\tau^\infty,i}\|_\infty, \dots, \|Q_T^{\tau^\infty,i}\|_\infty \right)_{i \in [N]} \rangle, \end{aligned}$$

where $\bar{\mathcal{S}} = \begin{bmatrix} \mathcal{T}_1 & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathcal{T}_2 & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathcal{T}_N \end{bmatrix}$, $\mathcal{T}_i = \begin{bmatrix} \beta_i^T & 0 & \dots & 0 \\ 0 & \beta_i^{T-1} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \beta_i \end{bmatrix}$, and $\mathbf{0}$ is a $(T-1) \times (T-1)$

dimensional matrix made of 0's. Here, the terms $\|Q_T^{\tau^\infty,i}\|_\infty$ are due to the fact that τ^∞ induces an infinite-horizon MFE, which can be directly obtained from Lemma 2 through recursive iterations as we did in the proof of Lemma 5.

Since $V(t_*) < 1$, we can rearrange the inequality above to obtain that

$$\begin{aligned} & \min_i c_i t_*^{T-t} \sum_{j \in [N]} \|\tau_{t,j}^\infty|_T - \tau_{t,j}^T\|_1 \leq \langle \hat{t}_*, (\|\tau_{t,j}^\infty|_T - \tau_{t,j}^T\|_1)_{t=1,j=1}^{T-1,N} \rangle \\ & \leq \frac{1}{1 - \rho(B(t_*))} \underbrace{\langle \hat{t}_*, \bar{\mathcal{S}} \left(\|Q_T^{\tau^\infty,i}\|_\infty, \|Q_T^{\tau^\infty,i}\|_\infty, \dots, \|Q_T^{\tau^\infty,i}\|_\infty \right)_{i \in [N]} \rangle}_{=I}. \end{aligned}$$

It remains to control the term I . First, note that, as $t_* \beta_{\arg \max_i \beta_i} < 1$, it holds that

$$I \leq N \max_i c_i \langle \hat{t}_*, \mathcal{T}_{\arg \max_i \beta_i} \left(\underbrace{\max_i \|Q_T^{\tau^\infty,i}\|_\infty, \max_i \|Q_T^{\tau^\infty,i}\|_\infty, \dots, \max_i \|Q_T^{\tau^\infty,i}\|_\infty}_{T-1 \text{ times}} \right) \rangle$$

$$\begin{aligned}
&\leq N \max_i c_i \max_i \|Q_T^{\tau^\infty, i}\|_\infty \beta_{\max} \sum_{j=1}^{T-1} \beta_{\max}^{T-j} t_*^{T-j} \\
&\leq N \max_i c_i \max_i \|Q_T^{\tau^\infty, i}\|_\infty \beta_{\max}^2 t_*^{\frac{\beta_{\max}^{T-1} t_*^{T-1} - 1}{\beta_{\max} t_* - 1}}.
\end{aligned}$$

Consequently, for the differences at section t throughout all the population groups, and using the fact that $c_i > 0$ for all $i \in [N]$, we obtain the following

$$\sum_{j \in [N]} \|\tau_{t,j}^\infty|_T - \tau_{t,j}^T\|_1 \leq \frac{N}{1 - \rho(B(t_*))} \frac{\max_i c_i}{\min_i c_i} \max_i \|Q_T^{\tau^\infty, i}\|_\infty \beta_{\max}^2 t_*^{t+1} \frac{\beta_{\max}^{T-1} t_* - \frac{1}{t_*^{T-t}}}{\beta_{\max} t_* - 1}.$$

Since $\beta_{\max} < 1$ and $t_* > 1$, as $T \rightarrow \infty$, we obtain that $\sum_{j \in [N]} \|\tau_{t,j}^\infty|_T - \tau_{t,j}^T\|_1 \rightarrow 0$. Furthermore, since $t_* \beta_{\max} < 1$, the dominating term in the sum above is $\frac{1}{t_*^{T-t}}$, which gives us the desired error bound.

It remains to prove the uniqueness of infinite-horizon non-stationary MFE under any given family of initial state-measures. For this purpose, we will demonstrate this for the initial state measure τ_0 , which does not break the generality. For each T , there exists a unique finite-horizon MFE as $\rho(\mathcal{Z}_\infty) < 1$. Furthermore, the error bound above shows that state-measures obtained under this MFE converge to any infinite-horizon MFE. From the uniqueness of the limit, it follows that all the infinite-horizon non-stationary MFE are obtained under a single state-measure flow τ^∞ . Lastly, one can use dynamical programming to induce a unique family of Q -functions under τ^∞ , see [18, Proposition 3.10]. Due to the existence of the regularizer term in our setting, there is a unique policy induced by these Q -functions, which implies that there is a unique infinite-horizon MFE. $\square$

In the single population case, we obtain the following formulation that is easier to handle:

Corollary 2. *Suppose $N = 1$, i.e. we are in the single population case. If there exists $t_0 > 1$ such that*

$$\frac{\bar{K}_1}{t_0} + \frac{\bar{L}_1 K_1}{\rho_1} \frac{1}{1 - \beta_1 t_0} < 1,$$

then under any initial state-measure $\mu_0 \in \mathcal{P}(X)$ there exists a unique infinite-horizon non-stationary MFE.

Proof. This follows directly from Theorem 6. $\square$

To obtain a rate of convergence between finite-horizon MFE and stationary MFE, next, we establish an asymptotic discount optimality criterion for infinite-horizon non-stationary MFE. First, we define an invariant space for $\mathcal{T}_{\lambda_\infty}$ using the relation $\lambda_\infty = \rho(B(r_*))$ for some r_* .

Lemma 24. *Suppose that $V(t_*) < 1$ holds for some $t_* > 1$. Let $v_i = c_i(1, 1/t_*, 1/t_*^2, \dots)$, where $(c_1, c_2, \dots, c_N)$ is a left positive Perron eigenvector of $B(t_*)$. Then, $\mathcal{T}_{\lambda_\infty}$ maps the set*

$$\mathcal{X}_{\lambda_\infty} = \left\{ (g_1, g_2, \dots, g_N) \in (\mathbb{R}^N)^N : \sup_{i \in [N]} \sup_{k \in \mathbb{N}} |g_i(k) v_i(k)| \leq 2|X| \right\}$$

into itself.

Proof. Since $\mathcal{T}_{\lambda_\infty}$ has a positive fixed point, Lemmas 19 and 20 imply that the following equality must hold

$$1 = \sum_{i \in [N]} \frac{K_i z_0 + \frac{\bar{L}_i K_i}{\rho_i} \frac{z_0}{z_0 - \beta_i}}{\lambda_\infty - (\bar{K}_i - K_i) z_0} \quad (46)$$

for some $z_0 \in (\beta_{\max}, \frac{\lambda_\infty}{\max_i (\bar{K}_i - K_i)})$.

Let $(c_1, c_2, \dots, c_N)$ be a positive left Perron eigenvector of $B(1/z_0)$. For any given $i \in [N]$, set $v_i = c_i(1, z_0, z_0^2, \dots)$. Define

$$\mathcal{X}_{\lambda_\infty} = \{(g_1, g_2, \dots, g_m) \in (\mathbb{R}^{\mathbb{N}})^N : \sup_{k \in \mathbb{N}} |g_i(k)v_i(k)| \leq 2|X|\},$$

where $z_0 = 1/t_*$ in this case. We claim that $\mathcal{T}_{\lambda_\infty}$ maps $\mathcal{X}_{\lambda_\infty}$ to itself. Indeed, for $k \geq 2$ and any $i \in [N]$ for any $g \in \mathcal{X}_{\lambda_\infty}$, equations (14) and (46) imply that

$$z_0^k (\mathcal{T}_\lambda(g))_i(k) \leq \frac{2|X|\rho(B(1/z_0))}{\lambda} = 2|X|.$$

For $k = 1$, (13) also implies the same bound. Therefore, $\mathcal{X}_{\lambda_\infty}$ is an invariant set under $\mathcal{T}_{\lambda_\infty}$. $\square$

Using the space construction above, we now state our asymptotic discount optimality result between infinite-horizon non-stationary MFE and stationary MFE.

Theorem 7. *Let $(\pi^\infty, \tau^\infty) \in \text{MFE}_{\tau_0}$ and (π, τ) be a stationary MFE. If there exists $t_* > 1$ such that $V(t_*) < 1$, then for all $k \geq 0$ we obtain*

$$\sup_{i \in [N], t \geq k} \|\tau_{t,i}^\infty - \tau_i\|_1 \leq O \left(\frac{\max_j (K_j + \frac{\bar{L}_j K_j}{\rho_j})}{1 - \rho(B(t_*))} \right)^k.$$

In particular, there exists a unique stationary MFE.

Proof. For any $x \in \mathcal{X}_{\lambda_\infty}$, let

$$\|x\| := \sup_{i \in [N]} \sup_{k \in \mathbb{N}} |x_i(k)v_i(k)|.$$

Note that $\|\cdot\|$ is a norm on $\mathcal{X}_{\lambda_\infty}$. For all $x \in \mathcal{X}_{\lambda_\infty}$, under the operator norm induced by $\|\cdot\|$, we obtain the following inequality

$$\|\lambda_\infty \mathcal{T}_{\lambda_\infty}\| = \sup_{x \in \mathcal{X}_{\lambda_\infty}} \frac{\|\lambda_\infty \mathcal{T}_{\lambda_\infty} x\|}{\|x\|} \leq \lambda_\infty \leq \rho(B(t_*)) < 1,$$

i.e. $\lambda_\infty \mathcal{T}_{\lambda_\infty}$ is a contraction on $(\mathcal{X}_{\lambda_\infty}, \|\cdot\|)$. Since $t_* > 1$, and that space of probability measures on X is universally bounded under the 1-norm, we have $(\|\tau_{k,i}^\infty - \tau_i\|_1)_{i,k} \in \mathcal{X}_{\lambda_\infty}$.

Let $(e_{i,k})_{i \in [N], k \geq 1} \in (\mathbb{R}^{\mathbb{N}})^N$ be the canonical basis. Then, it holds that

$$\begin{aligned} (\|\tau_{k,i}^\infty - \tau_i\|_1)_{i \in [N], k \geq m} &\leq \lambda_\infty \mathcal{T}_{\lambda_\infty} ((\|\tau_{k,i}^\infty - \tau_i\|_1)_{i,k}) + \sum_i e_{i,1} \otimes (K_i + \frac{\bar{L}_i K_i}{\rho_i}) \|\tau_{m-1,i}^\infty - \tau_i\|_1 \\ &\vdots \\ &\leq (I - \lambda_\infty \mathcal{T}_{\lambda_\infty})^{-1} \sum_i e_{i,1} \otimes (K_i + \frac{\bar{L}_i K_i}{\rho_i}) \|\tau_{m-1,i}^\infty - \tau_i\|_1, \end{aligned}$$

where $(I - \lambda_\infty \mathcal{T}_{\lambda_\infty})^{-1} = \sum_{k=0}^{\infty} (\lambda_\infty \mathcal{T}_{\lambda_\infty})^k$ is the Neumann series. Thus, we must have

$$\begin{aligned} \|(\|\tau_{k,i}^\infty - \tau_i\|_1)_{i \in [N], k \geq m}\| &\leq \frac{\max_j (K_j + \frac{\bar{L}_j K_j}{\rho_j})}{1 - \rho(B(t_*))} \max_i c_i \|\tau_{m-1,i}^\infty - \tau_i\|_1 \\ &\leq \frac{\max_j (K_j + \frac{\bar{L}_j K_j}{\rho_j})}{1 - \rho(B(t_*))} \|(\|\tau_{k,i}^\infty - \tau_i\|_1)_{i \in [N], k \geq (m-1)}\| \\ &\vdots \\ &\leq \left(\frac{\max_j (K_j + \frac{\bar{L}_j K_j}{\rho_j})}{1 - \rho(B(t_*))} \right)^m \max_i c_i \|\tau_{0,i}^\infty - \tau_i\|_1, \end{aligned}$$

where we have used that $\|(I - \lambda_\infty \mathcal{T}_{\lambda_\infty})^{-1}\| \leq \frac{1}{1 - \rho(B(t_*))}$. $\square$

A rate of convergence result between finite-horizon MFE and stationary MFE can be obtained directly from Theorems 6 and 7.

Corollary 3. *Let (π, τ) be a stationary MFE and $(\boldsymbol{\pi}^T, \boldsymbol{\tau}^T) \in \text{MFE}_{T, \tau_0}$. Then, we have that*

$$\|\tau_{k,i}^T - \tau_i\|_1 \leq O \left(\left(\frac{\max_j (K_1 + \frac{\bar{L}_i K_i}{\rho_i})}{1 - \rho(B(t_*))} \right)^k + \frac{1}{t_*^{T-k}} \right)$$

Proof. Follows from Theorems 6 and 7 after a straightforward triangle inequality, so we omit the details. $\square$

6 An Invariance Principle for Contraction under Slow-Fast Dynamic Updates for Lipschitz Systems

The quasi-static algorithm in Section 3 updates populations' Q -functions on a slower time scale than their state measures. As a result, the joint evolution of the Q -functions and state measures follows a slow-fast MFG dynamic. This naturally raises the question of whether the contraction condition established in the previous sections can be relaxed by allowing some populations to evolve on faster time scales than others.

For general dynamical systems, separating the dynamics into different time scales can improve convergence: quasi-static updates may converge to an equilibrium under conditions weaker than global contractivity of the full system; see [11, Remark 2]. In this section, however, we show that this effect does not arise when the dynamics are governed by a nonnegative matrix. In that case, introducing slow-fast dynamics does not relax the contractivity requirement, regardless of the quasi-static regime considered. Since our contraction results are obtained by majorizing the original system with a dynamical system of this type, it follows that these contraction conditions cannot be improved by allowing some populations to update their parameters faster than others.

Definition 4. *For each $i = 1, 2, \dots, N$, let (X_i, d_{X_i}) be a complete metric space. For each $j = 1, 2, \dots, N$, let*

$$f_j : \prod_{i=1}^N X_i \rightarrow X_j$$

be a map satisfying

$$d_{X_j}(f_j(x_1, \dots, x_N), f_j(\tilde{x}_1, \dots, \tilde{x}_N)) \leq \sum_{i=1}^N a_{j,i} d_{X_i}(x_i, \tilde{x}_i),$$

where $a_{j,i} \geq 0$ for all i, j .

We call the matrix $A = [a_{j,i}]_{j,i=1}^N$ the contraction matrix of the family $(f_j)_{j=1}^N$ on $\prod_{i=1}^N (X_i, d_{X_i})$.

Fix $0 < N_1 < N$. For any given $(x_j)_{j=N_1+1}^N \in \prod_{j=N_1+1}^N X_j$, we define a *quasi-static update* with respect to the variables $(x_{N_1+j})_{j=1}^{N-N_1}$ to be any element $(x_1, \dots, x_{N_1})[x_{N_1+1}, \dots, x_N] \in \prod_{i=1}^{N_1} X_i$ satisfying

$$\begin{aligned} & (f_1, \dots, f_{N_1}) \left((x_1, \dots, x_{N_1})[x_{N_1+1}, \dots, x_N], x_{N_1+1}, \dots, x_N \right) \\ &= (x_1, \dots, x_{N_1})[x_{N_1+1}, \dots, x_N]. \end{aligned}$$

Write

$$A = \begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix},$$

where $A_1 \in \mathbb{R}^{N_1 \times N_1}$ is the leading principal submatrix of A of order N_1 . For a given $(x_{N_1+j})_{j=1}^{N-N_1}$, the uniqueness of $(x_1, \dots, x_{N_1})[x_{N_1+1}, \dots, x_N]$ can be guaranteed by the condition $\rho(A_1) < 1$.

Assume now that $\rho(A_1) < 1$. Then the uniqueness of $(x_{N_1+1}, \dots, x_N)$ satisfying

$$(f_{N_1+1}, \dots, f_N)\left((x_1, \dots, x_{N_1})[x_{N_1+1}, \dots, x_N], x_{N_1+1}, \dots, x_N\right) = (x_{N_1+1}, \dots, x_N)$$

can be guaranteed by the condition $\rho(A_4 + A_3(I - A_1)^{-1}A_2) < 1$.

Thus, updating the first N_1 coordinates of the iteration of $(f_i)_{i \in [N]}$ and then updating the remaining $N - N_1$ coordinates yields a slow-fast dynamical update scheme for the family $(f_i)_{i \in [N]}$. In the next result, we show that, in general,

$$\rho(A) < 1 \iff \rho(A_1) < 1 \text{ and } \rho(A_4 + A_3(I - A_1)^{-1}A_2) < 1.$$

In particular, using only Lipschitz coefficients, one cannot establish convergence of the quasi-static updates of $(f_i)_{i=1}^N$ to an equilibrium unless one can already prove convergence of the direct iteration of the maps $(f_i)_{i=1}^N$.

Theorem 8. *Let $m \in \mathbb{N}$, let $n_1, \dots, n_m \in \mathbb{N}$ be arbitrary, and set $N := n_1 + \dots + n_m$. Let $A = [A_{ij}]_{i,j=1}^m \in \mathbb{R}_{\geq 0}^{N \times N}$, $A_{ij} \in \mathbb{R}^{n_i \times n_j}$. Thus A is a nonnegative matrix partitioned into $m \times m$ blocks with arbitrary block sizes, and each diagonal block A_{ii} is automatically square.*

Define recursively a sequence of nonnegative matrices $R_1, \dots, R_m$ as follows: $R_1 := A$, and for each $k = 1, \dots, m-1$, write R_k in the 2×2 block form

$$R_k = \begin{bmatrix} B_k & C_k \\ D_k & E_k \end{bmatrix},$$

where $B_k \in \mathbb{R}^{n_k \times n_k}$, $C_k \in \mathbb{R}^{n_k \times (n_{k+1} + \dots + n_m)}$, $D_k \in \mathbb{R}^{(n_{k+1} + \dots + n_m) \times n_k}$, $E_k \in \mathbb{R}^{(n_{k+1} + \dots + n_m) \times (n_{k+1} + \dots + n_m)}$, and define $R_{k+1} := E_k + D_k(I - B_k)^{-1}C_k$, whenever $(I - B_k)^{-1}$ exists. Then $\rho(A) < 1$ if and only if $\rho(B_k) < 1$ for all $k = 1, \dots, m-1$, and $\rho(R_m) < 1$.

Proof. We will proceed with induction on m . First consider the case $m = 2$. Then

$$A = \begin{bmatrix} B_1 & C_1 \\ D_1 & E_1 \end{bmatrix},$$

where $B_1 \in \mathbb{R}^{n_1 \times n_1}$, $C_1 \in \mathbb{R}^{n_1 \times n_2}$, $D_1 \in \mathbb{R}^{n_2 \times n_1}$, $E_1 \in \mathbb{R}^{n_2 \times n_2}$, and $R_2 = E_1 + D_1(I - B_1)^{-1}C_1$. We show that $\rho(A) < 1$ if and only if $\rho(B_1) < 1$ and $\rho(R_2) < 1$.

Assume first that $\rho(B_1) < 1$ and $\rho(R_2) < 1$. Since $B_1 \geq 0$ and $R_2 \geq 0$, the Neumann series imply

$$(I - B_1)^{-1} = \sum_{\ell=0}^{\infty} B_1^\ell \geq 0, \quad (I - R_2)^{-1} = \sum_{\ell=0}^{\infty} R_2^\ell \geq 0.$$

Now

$$I - A = \begin{bmatrix} I - B_1 & -C_1 \\ -D_1 & I - E_1 \end{bmatrix},$$

and the Schur complement of $I - B_1$ in $I - A$ is

$$(I - E_1) - D_1(I - B_1)^{-1}C_1 = I - R_2.$$

Hence $I - A$ is invertible, and the block inverse formula yields

$$(I - A)^{-1} = \begin{bmatrix} (I - B_1)^{-1} + (I - B_1)^{-1}C_1(I - R_2)^{-1}D_1(I - B_1)^{-1} & (I - B_1)^{-1}C_1(I - R_2)^{-1} \\ (I - R_2)^{-1}D_1(I - B_1)^{-1} & (I - R_2)^{-1} \end{bmatrix}.$$

Thus, as $(I - B_1)^{-1}, (I - R_2)^{-1} \geq 0$, it follows that $(I - A)^{-1} \geq 0$. Let $y \in \mathbb{R}^N$ be the vector of all ones and define $x := (I - A)^{-1}y$. Since $(I - A)^{-1}$ exists, $I - A$ has no non-zero rows. Then $x > 0$ and $(I - A)x = y > 0$, that is, $Ax = x - y < x$. Since $A \geq 0$, the existence of a vector $x > 0$ such that $Ax < x$ implies $\rho(A) < 1$.

Conversely, assume that $\rho(A) < 1$. Then $(I - A)^{-1} = \sum_{\ell=0}^{\infty} A^\ell \geq 0$. Let $x := (I - A)^{-1}y > 0$, and write $x = \begin{pmatrix} u \\ v \end{pmatrix}$ conformably with the block decomposition of A , where $u \in \mathbb{R}^{n_1}$, $v \in \mathbb{R}^{n_2}$. Then $Ax = x - y < x$, so $B_1u + C_1v < u$, $D_1u + E_1v < v$. In particular, $B_1u < u$. Since $B_1 \geq 0$, this implies $\rho(B_1) < 1$. Therefore $(I - B_1)^{-1} \geq 0$ exists, and from $(I - B_1)u > C_1v$ we obtain $u > (I - B_1)^{-1}C_1v$. Multiplying by $D_1 \geq 0$ gives $D_1u > D_1(I - B_1)^{-1}C_1v$. Using also $D_1u + E_1v < v$, we conclude that $(E_1 + D_1(I - B_1)^{-1}C_1)v < v$, that is, $R_2v < v$. Since $R_2 \geq 0$, this implies $\rho(R_2) < 1$. Thus the claim holds for $m = 2$.

Assume now that the theorem is true for all nonnegative block matrices with $(m-1) \times (m-1)$ block structure, and let A have $m \times m$ blocks. Write $A = R_1$ in the form $R_1 = \begin{bmatrix} B_1 & C_1 \\ D_1 & E_1 \end{bmatrix}$, where $B_1 \in \mathbb{R}^{n_1 \times n_1}$, $C_1 \in \mathbb{R}^{n_1 \times (n_2 + \dots + n_m)}$, $D_1 \in \mathbb{R}^{(n_2 + \dots + n_m) \times n_1}$, $E_1 \in \mathbb{R}^{(n_2 + \dots + n_m) \times (n_2 + \dots + n_m)}$. By the already proved $m = 2$ case, $\rho(R_1) < 1$ if and only if $\rho(B_1) < 1$ and $\rho(R_2) < 1$, where $R_2 = E_1 + D_1(I - B_1)^{-1}C_1$. Now R_2 is again a nonnegative matrix with an $(m-1) \times (m-1)$ block structure whose block sizes are $n_2, \dots, n_m$. Hence the induction hypothesis applies to R_2 and yields $\rho(R_2) < 1$ if and only if $\rho(B_k) < 1$ for $k = 2, \dots, m-1$, and $\rho(R_m) < 1$. Combining the two equivalences gives

$$\rho(A) = \rho(R_1) < 1 \iff \rho(B_1) < 1, \rho(B_2) < 1, \dots, \rho(B_{m-1}) < 1, \rho(R_m) < 1.$$

This completes the induction. $\square$

From the perspective of stochastic approximation, Theorem 8 shows that if the only available information about a system is its Lipschitz coefficients, then multi-time-scale stochastic approximation [8] cannot be shown to converge unless the system is already known to converge under single-time-scale updates.

We now discuss several applications of Theorem 8 to MFGs. In [4], for single-population MFGs, consecutive updates of the Q -function and the state measure are used to compute a stationary MFE when the only available information about the system consists of Lipschitz bounds on its components. Theorems 2 and 8 show that the same conclusion remains valid in the multi-population setting: if one only knows the Lipschitz coefficients of the system, then using consecutive updates of the Q -functions and state measures to compute a stationary MFE converges under the same conditions as the quasi-static approach.

Corollary 4. *Consider the map*

$$((Q_i)_{i=1}^N, \tau) \in \prod_{i=1}^N \mathcal{C}_i \times \prod_{i=1}^N \mathcal{P}(X) \mapsto ((H_{1,i}(Q_i, \tau))_{i=1}^N, (H_{2,i}(Q_i, \tau))_{i=1}^N).$$

Its contraction matrix on the product space $\prod_{i=1}^N (\mathcal{C}_i, \|\cdot\|_\infty) \times \prod_{i=1}^N (\mathcal{P}(X), \|\cdot\|_1)$ has spectral radius strictly less than 1 if and only if

$$\sum_{i \in [N]} \left(K_i + \frac{\bar{L}_i K_i}{\rho_i(1 - \beta_i)} \right) \frac{1}{1 - (\bar{K}_i - K_i)} < 1.$$

Proof. This result directly follows from Theorems 2 and 8 since the Q -functions are updated in a quasi-static manner in Theorem 2. $\square$

In the next corollary, we show that, in the regularized setting, if one uses only Lipschitz properties of the system, then allowing some populations to update their state measures faster than others does not yield a more relaxed contraction condition for the stationary MFG.

Corollary 5. *Suppose that $\{1, 2, \dots, N\} = \underbrace{\{1, 2, \dots, N_1\}}_{I_1} \cup \underbrace{\{N_1 + 1, \dots, N\}}_{I_2}$. For any family of*

state measures $\nu \in \prod_{j=N_1+1}^N \mathcal{P}(X)$, define $\tau(\nu) \in \prod_{j=1}^{N_1} \mathcal{P}(X)$ to be the family of state measures satisfying

$$\tau(\nu) = (H_{2,i}(Q^{(\tau(\nu), \nu)), i}, (\tau(\nu), \nu))_{i=1}^{N_1}.$$

Then the contraction matrix of the map

$$\boldsymbol{\tau} \mapsto (H_{2,i}(Q^{(\boldsymbol{\tau}, \boldsymbol{\nu})}, i, (\boldsymbol{\tau}, \boldsymbol{\nu})))_{i=1}^{N_1}$$

on $\prod_{j=1}^{N_1}(\mathcal{P}(X), \|\cdot\|_1)$, and the contraction matrix of the map

$$\boldsymbol{\nu} \mapsto (H_{2,i}(Q^{(\boldsymbol{\tau}(\boldsymbol{\nu}), \boldsymbol{\nu})}, i, (\boldsymbol{\tau}(\boldsymbol{\nu}), \boldsymbol{\nu})))_{i=N_1+1}^N$$

on $\prod_{j=N_1+1}^N(\mathcal{P}(X), \|\cdot\|_1)$, both have spectral radius less than 1 if and only if

$$\sum_{i \in [N]} \left(K_i + \frac{\bar{L}_i K_i}{\rho_i(1 - \beta_i)} \right) \frac{1}{1 - (\bar{K}_i - K_i)} < 1.$$

Proof. This result directly follows from Theorems 2 and 8. $\square$

It follows directly from Lemma 5 that, in the multi-population finite-horizon setting, verifying contractivity of quasi-static updates with different time scales is already nontrivial even with only two populations. Indeed, one must check the spectral radius of

$$T_2(\bar{K}_2) + T_2(K_2)(I - T_1(\bar{K}_1))^{-1}T_1(K_1),$$

which is more involved than directly controlling the matrix $\mathcal{S}_T$, due to the presence of the inverse term $(I - T_1(\bar{K}_1))^{-1}$.

Another consequence of Theorem 8 is that, even in the finite-horizon setting, the contraction criterion of Theorem 4 cannot be sharpened using only the Lipschitz properties of the system. In particular, introducing heterogeneous update time scales across populations does not lead to stronger convergence guarantees.

Corollary 6. Suppose that $\{1, 2, \dots, N\} = \underbrace{\{1, 2, \dots, N_1\}}_{I_1} \cup \underbrace{\{N_1 + 1, N_1 + 2, \dots, N\}}_{I_2}$. For any

$\boldsymbol{\nu}^T = (\nu_{t,i})_{t=0, i=N_1+1}^{T-1, N} \in \prod_{i=N_1+1}^N \{\nu_{0,i}\} \times \prod_{t=1}^{T-1} \prod_{i=N_1+1}^N \mathcal{P}(X)$, let $\boldsymbol{\tau}^T(\boldsymbol{\nu}^T) \in \prod_{i=1}^{N_1} \{\tau_{0,i}\} \times \prod_{t=1}^{T-1} \prod_{i=1}^{N_1} \mathcal{P}(X)$ denote the family of state measures defined by

$$\boldsymbol{\tau}^T(\boldsymbol{\nu}^T) = \left((\tau_{0,i})_{i=1}^{N_1}, \left(H_{2,i}(Q_t^{(\boldsymbol{\tau}(\boldsymbol{\nu}), \boldsymbol{\nu})}, i, (\boldsymbol{\tau}(\boldsymbol{\nu}), \boldsymbol{\nu}_t)) \right)_{i=1, t=1}^{N_1, T-1} \right).$$

Then the contraction matrices of the maps

$$\boldsymbol{\tau} \mapsto \left((\tau_{0,i})_{i=1}^{N_1}, \left(H_{2,i}(Q_t^{(\boldsymbol{\tau}, \boldsymbol{\nu})}, i, (\boldsymbol{\tau}, \boldsymbol{\nu}_t)) \right)_{i=1, t=1}^{N_1, T-1} \right)$$

and

$$\boldsymbol{\nu} \mapsto \left((\nu_{0,i})_{i=N_1+1}^N, \left(H_{2,i}(Q_t^{(\boldsymbol{\tau}(\boldsymbol{\nu}), \boldsymbol{\nu})}, i, (\boldsymbol{\tau}(\boldsymbol{\nu}), \boldsymbol{\nu}_t)) \right)_{i=N_1+1, t=1}^{N, T-1} \right)$$

on $\prod_{i=1}^{N_1} \{\tau_{0,i}\} \times \prod_{t=1}^{T-1} \prod_{i=1}^{N_1}(\mathcal{P}(X), \|\cdot\|_1)$ and $\prod_{i=N_1+1}^N \{\nu_{0,i}\} \times \prod_{t=1}^{T-1} \prod_{i=N_1+1}^N(\mathcal{P}(X), \|\cdot\|_1)$, respectively, have spectral radius less than 1 for all $T \geq 0$ if and only if

$$\inf_{\bar{K}_\infty < r < \beta_{\max}^{-1}} \sum_{i \in [N]} \frac{\frac{K_i + a_i}{r} + \frac{a_i \beta_i}{1 - \beta_i r}}{1 - \frac{\bar{K}_i - K_i}{r}} < 1.$$

Proof. This result directly follows from Theorems 2 and 4. $\square$

7 Conclusion

In this work, we have presented contraction conditions for multi-population discrete-time regularized MFGs. Using the variational nature of these contraction conditions, we have obtained error bounds between finite-horizon and infinite-horizon MFE. Furthermore, we have shown that these contraction conditions cannot be improved by introducing slow-fast dynamics when the only knowledge about the system is the Lipschitz coefficients.

The variational contraction conditions we have presented capture the asymptotics of the spectral radius that correspond to the contraction matrix in our setting, and provides more general working conditions under which one can obtain error bounds between finite-horizon and infinite-horizon MFE compared to the one given in [6]. However, our method does not provide a tight lower bound for the spectral radius in the finite-horizon setting unlike [6, Theorem 2]. Our numerical results suggest that the spectral radius in the multi-population case should admit a lower bound similar to the one derived in [6]. Establishing a tight lower bound for $\rho(\mathcal{S}_T)$ is an important direction for future research, as it would provide further insight into the asymptotic behavior of contraction rates for finite-horizon MFE in the multi-population setting.

References

- [1] Shubham Aggarwal et al. “Semantic Communication in Multiteam Dynamic Games: A Mean Field Perspective”. In: *IEEE Transactions on Automatic Control* 71.1 (2026), pp. 49–64.
- [2] Lars Valerian Ahlfors. *Complex Analysis*. Vol. 3. McGraw-Hill New York, 1979.
- [3] Berkay Anahtarci, Can Deha Kariksiz, and Naci Saldi. “Q-learning in regularized mean-field games”. In: *Dynamic Games and Applications* 13.1 (2023), pp. 89–117.
- [4] Berkay Anahtarci, Can Deha Kariksiz, and Naci Saldi. “Value iteration algorithm for mean-field games”. In: *Systems & Control Letters* 143 (2020), p. 104744.
- [5] Andrea Angiuli et al. “Analysis of multiscale reinforcement Q-learning algorithms for mean field control games”. In: *Applied Mathematics & Optimization* 93.2 (2026), p. 27.
- [6] Uğur Aydın, Tamer Başar, and Naci Saldi. “Approximation of Discrete-Time Infinite-Horizon Mean-Field Equilibria via Finite-Horizon Mean-Field Equilibria”. In: *arXiv preprint arXiv:2509.01039* (2025).
- [7] Uğur Aydın, Tamer Başar, and Naci Saldi. “Approximation of discrete-time infinite-horizon mean-field equilibria via finite-horizon mean-field equilibria”. In: *2025 IEEE 64th Conference on Decision and Control (CDC)*. IEEE. 2025, pp. 305–310.
- [8] Vivek S Borkar. “Stochastic approximation with two time scales”. In: *Systems & Control Letters* 29.5 (1997), pp. 291–294.
- [9] Vivek S Borkar and Sean P Meyn. “The ODE method for convergence of stochastic approximation and reinforcement learning”. In: *SIAM Journal on Control and Optimization* 38.2 (2000), pp. 447–469.
- [10] René Carmona, Ludovic Tangpi, and Kaiwen Zhang. “A probabilistic approach to discounted infinite horizon and invariant mean field games”. In: *arXiv preprint arXiv:2407.03642* (2024).
- [11] T. T. Doan. “Fast Nonlinear Two-Time-Scale Stochastic Approximation: Achieving $\mathcal{O}(1/k)$ Finite-Sample Complexity”. In: *IEEE Transactions on Automatic Control* 71.1 (2026), pp. 230–242.
- [12] Philippe Flajolet and Robert Sedgewick. *Analytic Combinatorics*. Cambridge University Press, 2009.
- [13] Xin Guo et al. “Learning mean-field games”. In: *Advances in Neural Information Processing Systems* 32 (2019).
- [14] Roger A Horn and Charles R Johnson. *Matrix Analysis*. Cambridge University Press, 2012.

- [15] Jiawei Huang, Batuhan Yardim, and Niao He. “On the statistical efficiency of mean-field reinforcement learning with general function approximation”. In: *International Conference on Artificial Intelligence and Statistics*. PMLR. 2024, pp. 289–297.
- [16] Leonid (Aryeh) Kontorovich and Kavita Ramanan. “Concentration inequalities for dependent random variables via the martingale method”. In: *The Annals of Probability* 36.6 (2008), pp. 2126–2158. DOI: 10.1214/07-AOP384. URL: <https://doi.org/10.1214/07-AOP384>.
- [17] Huaning Liu et al. “Incorporating Authority Perception, Economic Status, and Behavioral Response in Infectious Disease Control”. In: *arXiv preprint arXiv:2512.23188* (2025).
- [18] N. Saldi, T. Başar, and M. Raginsky. “Markov–Nash equilibria in mean-field games with discounted cost”. In: *SIAM Journal on Control and Optimization* 56.6 (2018), pp. 4256–4287.
- [19] Barna Pásztor, Andreas Krause, and Ilija Bogunovic. “Efficient model-based multi-agent mean-field reinforcement learning”. In: *Transactions on Machine Learning Research* (2023).
- [20] Piotr Wićcek. “Multiple-population discrete-time mean field games with discounted and total payoffs: The existence of equilibria”. In: *Dynamic Games and Applications* 14.4 (2024), pp. 997–1026.
- [21] Liping Xu et al. “Q-learning-driven multi-population cooperative evolutionary algorithm with local search for scheduling of network-shared manufacturing resources”. In: *Computers & Operations Research* 180 (2025), p. 107076.
- [22] Muhammad Aneeq uz Zaman, Erik Miehl, and Tamer Başar. “Reinforcement learning for non-stationary discrete-time linear–quadratic mean-field games in multiple populations”. In: *Dynamic Games and Applications* 13.1 (2023), pp. 118–164.

A Proof of Lemma 2

Proof of Lemma 2. First, we will prove the inequality (2). Using the triangle inequality, we obtain that

$$\begin{aligned}
& \|H_{1,i}(Q^1, \tau) - H_{1,i}(Q^2, \tilde{\tau})\|_{\infty} \\
&= \sup_{(x,u) \in X \times \mathcal{P}(A)} \left| C_i(x, u, \tau_1, \dots, \tau_N) - C_i(x, u, \tilde{\tau}_1, \dots, \tilde{\tau}_N) \right. \\
&\quad \left. + \beta \sum_{y \in X} Q_{\min}^1(y) P_i(y|x, u, \tau_1, \dots, \tau_N) - \beta \sum_{y \in X} Q_{\min}^2(y) P_i(y|x, u, \tilde{\tau}_1, \dots, \tilde{\tau}_N) \right| \\
&\leq \sup_{(x,u) \in X \times \mathcal{P}(A)} |C_i(x, u, \tau_1, \dots, \tau_N) - C_i(x, u, \tilde{\tau}_1, \dots, \tilde{\tau}_N)| \\
&\quad + \sup_{(x,u) \in X \times \mathcal{P}(A)} \beta_i \underbrace{\left| \sum_{y \in X} Q_{\min}^1(y) P_i(y|x, a, \tau_1, \dots, \tau_N) - \sum_{y \in X} Q_{\min}^2(y) P_i(y|x, a, \tilde{\tau}_1, \dots, \tilde{\tau}_N) \right|}_{(*)}.
\end{aligned}$$

We further bound the term $(*)$ as follows:

$$\begin{aligned}
& \left| \sum_{y \in X} Q_{\min}^1(y) P_i(y|x, a, \tau_1, \dots, \tau_N) - \sum_{y \in X} Q_{\min}^2(y) P_i(y|x, a, \tilde{\tau}_1, \dots, \tilde{\tau}_N) \right| \\
&\leq \left| \sum_{y \in X} Q_{\min}^1(y) (P_i(y|x, a, \tau_1, \dots, \tau_N) - P_i(y|x, a, \tilde{\tau}_1, \dots, \tilde{\tau}_N)) \right|
\end{aligned}$$

$$+ \left| \sum_{y \in X} (Q_{\min}^1 - Q_{\min}^2)(y) P_i(y|x, u, \tilde{\tau}_1, \dots, \tilde{\tau}_N) \right|.$$

Note that

$$\left| \sum_{y \in X} (Q_{\min}^1 - Q_{\min}^2)(y) P_i(y|x, u, \tilde{\tau}_1, \dots, \tilde{\tau}_N) \right| \leq \|Q_{\min}^1 - Q_{\min}^2\|_{\infty}.$$

Let $\|Q_{\min}^1\|_{\text{Lip}}$ be the smallest Lipschitz coefficient of $Q_{\min}^1$ over X . Using the concentration inequality [16, Lemma A.2], we directly obtain

$$\begin{aligned} & \left| \sum_{y \in X} Q_{\min}^1(y) (P_i(y|x, a, \tau_1, \dots, \tau_N) - P_i(y|x, a, \tilde{\tau}_1, \dots, \tilde{\tau}_N)) \right| \\ & \leq \frac{\|Q_{\min}^1\|_{\text{Lip}}}{2} \|P_i(\cdot|x, a, \tau_1, \dots, \tau_N) - P_i(\cdot|x, a, \tilde{\tau}_1, \dots, \tilde{\tau}_N)\|_1 \\ & \leq \frac{K_i}{2} \|Q_{\min}^1\|_{\text{Lip}} \sum_{j \in [N]} \|\tau_j - \tilde{\tau}_j\|_1, \end{aligned}$$

where the last line is due to Assumption 1. Thus, one obtains

$$\begin{aligned} & \left| \sum_{y \in X} Q_{\min}^1(y) (P_i(y|x, a, \tau_1, \dots, \tau_N) - P_i(y|x, a, \tilde{\tau}_1, \dots, \tilde{\tau}_N)) \right| \\ & + \left| \sum_{y \in X} (Q_{\min}^1 - Q_{\min}^2)(y) P_i(y|x, u, \tilde{\tau}_1, \dots, \tilde{\tau}_N) \right| \\ & \leq \frac{K_i}{2} \|Q_{\min}^1\|_{\text{Lip}} \sum_{j \in [N]} \|\tau_j - \tilde{\tau}_j\|_1 + \|Q_{\min}^1 - Q_{\min}^2\|_{\infty}, \end{aligned}$$

It then follows that

$$\begin{aligned} & \sup_{(x, u) \in X \times \mathcal{P}(A)} \beta_i \left| \sum_{y \in X} Q_{\min}^1(y) P_i(y|x, a, \tau_1, \dots, \tau_N) - \sum_{y \in X} Q_{\min}^2(y) P_i(y|x, a, \tilde{\tau}_1, \dots, \tilde{\tau}_N) \right| \\ & \leq \sup_{(x, u) \in X \times \mathcal{P}(A)} \beta_i K_i \|Q_{\min}^1\|_{\text{Lip}} \sum_{j \in [N]} \|\tau_j - \tilde{\tau}_j\|_1 + \beta_i \|Q_{\min}^1 - Q_{\min}^2\|_{\infty} \\ & \leq \frac{K_i \beta_i}{2} \|Q_{\min}^1\|_{\text{Lip}} \sum_{j \in [N]} \|\tau_j - \tilde{\tau}_j\|_1 + \beta_i \|Q_{\min}^1 - Q_{\min}^2\|_{\infty}. \end{aligned}$$

Combining all the terms above, we then obtain

$$\|H_{1,i}(Q^1, \tau) - H_{1,i}(Q^2, \tilde{\tau})\|_{\infty} \leq \left(L_i + \frac{K_i \beta_i}{2} \|Q_{\min}^1\|_{\text{Lip}} \right) \sum_{j \in [N]} \|\tau_j - \tilde{\tau}_j\|_1 + \beta_i \|Q_{\min}^1 - Q_{\min}^2\|_{\infty}.$$

It remains to calculate $\|Q_{\min}^1\|_{\text{Lip}}$. First, note that $\|Q_{\min}^1\|_{\text{Lip}} \leq \sup_{u \in \mathcal{P}(A)} \sup_{x \neq y} |Q^1(x, u) - Q^1(y, u)|$. Since $Q^1(x, u) = \sum_{a \in A} \mathbf{q}^1(x, a) u(a) + \rho_i \Omega_i$ for some $\mathbf{q}^1 \in \tilde{\mathcal{C}}_i$, directly from the definition of $\mathbf{q}^1$ we obtain

$$\sup_{u \in \mathcal{P}(A)} |Q^1(x, u) - Q^1(y, u)| \leq \sup_{a \in A} \|\mathbf{q}^1(x, \cdot)\|_{\text{Lip}} \leq \frac{L_i}{1 - \frac{\beta_i K_i}{2}},$$

from which (2) follows.

For the rest of the proof, we will focus on showing that (3) holds. We start with the triangle inequality:

$$\begin{aligned}
& \|H_{2,i}(Q^1, \tau) - H_{2,i}(Q^2, \tilde{\tau})\|_1 \\
&= \sum_y \left| \sum_x P_i(y|x, \operatorname{argmin}_{u \in \mathcal{P}(A)} Q^1(x, u), \tau_1, \dots, \tau_N) \tau_i(x) - \sum_x P_i(y|x, \operatorname{argmin}_{u \in \mathcal{P}(A)} Q^2(x, u), \tilde{\tau}_1, \dots, \tilde{\tau}_N) \tilde{\tau}_i(x) \right| \\
&\leq \sum_y \left| \sum_x P(y|x, \operatorname{argmin}_{u \in \mathcal{P}(A)} Q^1(x, u), \tau_1, \dots, \tau_N) \tau_i(x) - \sum_x P_i(y|x, \operatorname{argmin}_{u \in \mathcal{P}(A)} Q^2(x, u), \tilde{\tau}_1, \dots, \tilde{\tau}_N) \tau_i(x) \right| \\
&\quad + \sum_y \left| \sum_x P_i(y|x, \operatorname{argmin}_{u \in \mathcal{P}(A)} Q^2(x, u), \tilde{\tau}_1, \dots, \tilde{\tau}_N) \tau_i(x) - \sum_x P_i(y|x, \operatorname{argmin}_{u \in \mathcal{P}(A)} Q^2(x, u), \tilde{\tau}_1, \dots, \tilde{\tau}_N) \tilde{\tau}_i(x) \right|.
\end{aligned}$$

Consequently, using Assumption 1, we must have

$$\begin{aligned}
& \|H_{2,i}(Q^1, \tau_1, \dots, \tau_N) - H_{2,i}(Q^2, \tilde{\tau}_1, \dots, \tilde{\tau}_N)\|_1 \\
&\leq \sum_x \left\| P_i(\cdot|x, \operatorname{argmin}_{u \in \mathcal{P}(A)} Q^1(x, u), \tau_1, \dots, \tau_N) - P_i(\cdot|x, \operatorname{argmin}_{u \in \mathcal{P}(A)} Q^2(x, u), \tilde{\tau}_1, \dots, \tilde{\tau}_N) \right\|_1 \tau_i(x) \\
&\quad + \frac{K_i}{2} \left(1 + \frac{\bar{L}_i}{\rho_i} \right) \|\tau_i - \tilde{\tau}_i\|_1 \\
&\leq K_1 \left(\sup_x \left\| \operatorname{argmin}_{u \in \mathcal{P}(A)} Q^1(x, u) - \operatorname{argmin}_{u \in \mathcal{P}(A)} Q^2(x, u) \right\|_1 + \sum_{j \in [N]} \|\tau_j - \tilde{\tau}_j\|_1 \right) \\
&\quad + \frac{K_i}{2} \left(1 + \frac{\bar{L}_i}{\rho_i} \right) \|\tau_i - \tilde{\tau}_i\|_1 \\
&\leq \frac{K_i}{\rho_i} \|Q^1 - Q^2\|_\infty + \bar{K}_i \|\tau_i - \tilde{\tau}_i\|_1 + K_i \sum_{j \neq i} \|\tau_j - \tilde{\tau}_j\|_1,
\end{aligned}$$

where on the last line we have used that

$$\sup_x \left\| \operatorname{argmin}_{u \in \mathcal{P}(A)} Q^1(x, u) - \operatorname{argmin}_{u \in \mathcal{P}(A)} Q^2(x, u) \right\|_1 \leq \frac{1}{\rho_i} \|Q^1 - Q^2\|_\infty;$$

see [3, Proposition 3]. In particular, we have obtained (3). $\square$